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Research paper

A wave-generation boundary deficit dominates a multi-event assessment of 1D XBeach non-hydrostatic

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ABSTRACT

In this work, a 1D XBeach non-hydrostatic model was assessed against 14 years of observations at the Field Research Facility, Duck, North Carolina, using 52 events stratified by energy class and sandbar state. The principal numerical finding is a systematic boundary transmission deficit. The 1D JONSWAP wave-generation boundary delivers on average 57% of the prescribed wave height, and the shortfall depends on the sign of the wave approach angle rather than its magnitude. A flat-bottom experiment that removes bathymetry, shoaling, and breaking reproduces the shortfall in full, identifying it as a property of the wave-generation boundary and not of the cross-shore physics. Referenced to the prescribed boundary height, model-observation correlation is indistinguishable from zero, but this measures boundary delivery and transformation together. Referenced to the energy actually delivered, a weak but resolvable correlation appears (0.21, 95% CI 0.11 to 0.30). Re-running the events with the boundary inflated to deliver the intended energy raises it to 0.30; predicting that inflation from wave direction alone and scoring only withheld events gives the same recovery, so it does not depend on hindsight. The assessment is diagnostic, and the seven most energetic events cannot be corrected in 1D.

1. Introduction

Wave energy dissipation by depth-induced breaking controls the transformation of incident waves from intermediate water depths to the shoreline (Battjes and Janssen, 1978; Thornton and Guza, 1983). The cross-shore structure of dissipation governs wave setup, infragravity wave generation, surf zone circulation, runup, and sediment transport, with direct consequences for coastal flood hazard assessment and shoreline protection design (Stockdon et al., 2006). Phase-resolving non-hydrostatic shallow-water models have emerged as a practical class of tools for simulating these processes at the individual wave scale. SWASH (Zijlema et al., 2011) and XBeach-NH (Roelvink et al., 2009; Smit et al., 2013) both apply a non-hydrostatic pressure correction to the depth-averaged nonlinear shallow-water equations to extend their applicability into the dispersive wave transformation regime. SWASH has been extensively assessed for wave transformation, runup, overtopping, and currents in laboratory and field settings (Smit et al., 2014; Henderson et al., 2022; Al Khalili et al., 2025; Rijnsdorp et al., 2014, 2015). XBeach-NH is embedded within the broader XBeach hydro-morphodynamic coastal impact framework (Roelvink et al., 2009), whose capabilities include intra-wave bed evolution, storm overwash, dune erosion, and sediment transport. In the present study, bed updating is disabled and XBeach is run in fixed-bed non-hydrostatic mode, allowing direct comparison against observed bathymetric states from CRAB surveys without coupling hydrodynamic error to uncertain sediment transport closure. The model has been assessed against runup and overtopping on laboratory profiles (Roelvink et al., 2018), and on fringing reef environments (Lashley et al., 2018). Field-scale assessments of XBeach-NH cross-shore transformation over barred sandy profiles, and systematic multi-event assessments in particular, remain comparatively few; this is the gap the present study addresses. In fact, long term archives also require explicit treatment of data provenance, quality control, sampling structure, and statistical dependence when used for broad inference (Biondi, 2026).

The dissipation process depends on the interaction between incident wave energy and the cross-shore bathymetric profile, and in particular the position, height, and geometry of nearshore sandbars. Two parameter combinations that produce nearly identical H_{m0} at individual gauge positions can yield substantially different cross-shore dissipation gradients, wave setup magnitudes, and infragravity wave energies. This degeneracy is not resolved by point-gauge comparisons alone. Calibration based on sparse gauge records tends to compensate for structural errors in the breaking parameterization through adjustments to friction or other free parameters (Ruessink et al., 2003), producing models that

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perform well in terms of bulk wave height statistics at the calibration site while leaving the underlying dissipation structure untested. A natural way to address this is to derive observed transformation and dissipation profiles from the wave energy flux gradient measured between adjacent gauges. The dissipation rate integrated over a cross-shore segment can be expressed as the negative derivative of the wave energy flux, $D(x) = -d(EC_g)/dx$, where E is the wave energy density and C_g is the group velocity (Thornton and Guza, 1983; Elgar et al., 1997). This formulation requires steady-state conditions, shore-normal propagation, and alongshore uniformity over the gauge array, and it assumes that bottom friction losses are small relative to breaking dissipation. When applied to a sufficiently dense array, it yields a discrete profile of depth-integrated dissipation and normalized wave height transformation that can be compared directly against model output. Normalizing wave heights by an offshore reference is a standard practice in field transformation studies (Ruessink et al., 2003; Apotsos et al., 2008).

Field-scale multi-event assessments of phase-resolving wave transformation models over barred sandy profiles remain rare. Prior systematic work at the FRF has focused on phase-averaged spectral wave models run continuously against the cross-shore gauge array (Smith et al., 2017), evaluating skill through integrated wave height statistics without assessing dissipation structure or phase-resolving fidelity. Earlier field campaigns that documented wave transformation and dissipation at the FRF and comparable sites (Thornton and Guza, 1983; Elgar et al., 1997; Baldock et al., 1998; Ruessink et al., 2001) were limited to single campaign windows and temporary dense arrays, precluding assessment across a wide range of wave energy conditions and sandbar morphological states. Furthermore, no reproducible multi-event comparison workflow has been established for 1D XBeach-NH at the FRF. The fidelity of boundary condition energy transmission in phase-resolving nearshore models has received limited systematic attention (de Ridder et al., 2021), and single-layer non-hydrostatic implementations face known dispersion limits in intermediate water that are relevant to the energetic events in this record. Similar combined numerical and experimental approaches have been used to quantify wave transformation, dissipation, and wave induced loading in coastal engineering applications (Biondi et al., 2026). Post-run analysis in the present study revealed a systematic, direction-dependent energy transmission deficit at the 1D model boundary, which precludes absolute H_{m0} comparison and motivates the normalized wave height transformation framework adopted as the primary assessment metric. Because both the normalized metric and the event stratification were finalized after inspecting model behaviour rather than pre-specified, the skill assessments presented here are explicitly characterized as exploratory diagnostic findings rather than confirmatory tests.

The FRF at Duck, North Carolina (Fig. 1), provides an exceptional opportunity to address these gaps. The FRF has maintained a continuous cross-shore array of wave gauges spanning from 26 m depth to the inner surf zone since the late 1970s, with instruments including directional Waverider buoys, Acoustic Wave and Current profilers (AWACs), and Paros pressure gauges at up to seven cross-shore positions. Co-located bathymetric surveys using the Coastal Research Amphibious Buggy (CRAB) have been conducted approximately monthly for over three decades, providing a detailed record of sandbar evolution. All data are freely accessible via the FRF CHL THREDDS server. The FRF record enables a multi-event model-data assessment over a wide range of forcing conditions and morphological states using a consistent instrumentation network that no purpose-designed field campaign could replicate.

This paper presents a 52-event field-scale assessment of XBeach-NH cross-shore wave transformation at Duck, North Carolina, using the transformation ratio $K = H_{m0}/H_{m0}^{BC}$ as the primary metric and the energy retention fraction as a secondary metric. The study has four objectives: (1) characterize the energy transmitted by the 1D JONSWAP boundary as a function of wave direction, energy class, and period; (2) document observed cross shore wave transformation across 52 events spanning 14 years, three energy classes, and different sandbar states; (3) assess fixed bed 1D XBeach NH against these observations using consistent processing and no event specific calibration; and (4) quantify sensitivity to grid resolution, spin up duration, and breaking steepness.

The analysis identifies a systematic deficit in the energy delivered by the 1D wave generation boundary. A flat bottom diagnostic isolates this deficit from bathymetric transformation, and additional tests examine its dependence on direction, depth, period, and numerical formulation. Because the deficit affects absolute model observation comparisons, boundary referenced results represent the combined effects of boundary transmission and cross shore transformation. Transformation skill is therefore evaluated separately using the energy delivered by the model and through corrected boundary simulations. The 52-event observational dataset provides an independent characterization of cross shore wave transformation at the FRF and supports practical

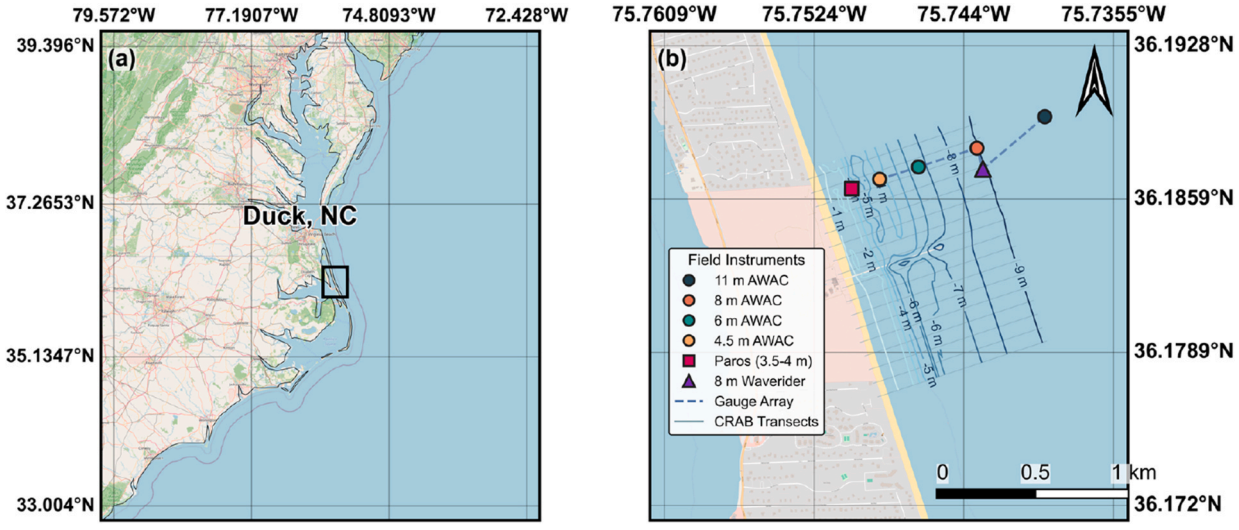


Fig. 1. Study site and field instrumentation at the U.S. Army Corps of Engineers Field Research Facility (FRF), Duck, North Carolina, USA. (a) Regional map showing the location of Duck on the Outer Banks. (b) Local map of the FRF nearshore observation array, including bathymetric contours from the CRAB survey DEM, CRAB transect lines, and the cross-shore wave gauge array comprising the 11, 8, 6, and 4.5 m AWACs, the Paros gauge in approximately 3.5-4 m water depth, and the 8 m Waverider buoy. Contours are referenced to local bathymetric elevation, and coordinates are given in geographic latitude and longitude.

recommendations for applying and interpreting 1D XBeach NH at barred sandy coasts with oblique wave climates.

The paper is organized as follows. Section 2 describes the study site, the FRF dataset, the event selection procedure, the methods for deriving observed transformation profiles, the XBeach-NH model configuration, the boundary condition energy deficit, and the comparison metrics. Section 3 presents the results. Section 4 discusses the implications for nearshore modeling practice. Section 5 provides the conclusions.

2. Methods

This study applies XBeach-NH in 1D mode as a deliberate scope decision: the objective is to document the performance envelope and boundary condition behavior of the 1D configuration at a field site with an oblique wave climate, rather than to optimize model performance.

2.1. Study area

The FRF is located on the Outer Banks of North Carolina at Duck (36.18°N, 75.75°W), a microtidal barrier island beach exposed to open Atlantic swell and nor'easter storm waves. The site has been continuously instrumented since the late 1970s and provides one of the longest and most comprehensive nearshore observation records in the world. The cross-shore instrument array used in this study spans from 26 m depth to the inner surf zone. It comprises a directional Waverider buoy at approximately 8 m depth, Acoustic Wave and Current profilers (AWACs) at nominal depths of 11 m, 8 m, 6 m, and 4.5 m, and a Paros pressure gauge array at approximately 3.5 to 4 m depth (Fig. 2). Co-located bathymetric surveys using the Coastal Research Amphibious Buggy (CRAB; Birkemeier and Mason, 1984) are conducted approximately monthly along shore-normal transects spanning y_{FRF} = 200 to 900 m, providing a long-term record of sandbar evolution. The FRF wave climate has a moderate annual mean H_{m0} of approximately 0.9 m, with energetic nor'easter events regularly exceeding 3 m from October to March and calm summer periods with H_{m0} frequently below 0.5 m (Fig. 4). All observational data used in this study are freely accessible via the USACE CHL THREDDS server without restriction.

2.2. Derivation of observed cross-shore wave transformation and dissipation profiles

The observed cross-shore dissipation rate is derived from the wave energy flux measured at adjacent gauge positions along the FRF cross-shore array. Under the assumption of shore-normal wave propagation and alongshore uniformity, the latter supported at this site by the alongshore-uniform nearshore circulation documented at the FRF (Feddersen and Guza, 2003), the depth-integrated wave energy balance in the cross-shore direction reduces to:

$$\frac{d(EC_g)}{dx} = -D(x) \tag{1}$$

where E is the depth-integrated wave energy density (J m^{-2}), C_g is the wave group velocity (m s^{-1}), x is the cross-shore coordinate positive onshore, and $D(x)$ is the total depth-integrated dissipation rate (W m^{-2}) due to wave breaking and bed friction. The wave energy density is computed from the variance density spectrum $S(f)$ measured at each gauge:

$$E = \rho g \int_{f_{\min}}^{f_{\max}} S(f)df = \frac{1}{8}\rho g H_{m0}^2 \tag{2}$$

where ρ is the water density (1025 kg m^{-3}), g is gravitational acceleration (9.81 m s^{-2}), and $H_{m0} = 4\sqrt{m_0}$ is the spectral significant wave height computed from the zeroth spectral moment m_0. The integration limits are set to $f_{\min} = 0.04$ Hz and $f_{\max} = 0.33$ Hz to match the frequency range of the FRF AWAC spectral products, which are archived from 0.04 to 0.50 Hz. Infragravity energy (typically $f < 0.04$ Hz at Duck; Elgar et al., 1992) is therefore excluded from all sea-swell H_{m0} and EC_g computations. This exclusion applies to both observed and modeled quantities and is consistent for the K and Φ comparisons. An infragravity-band assessment, which would be the primary motivation for non-hydrostatic over hydrostatic physics and is an important direction for future work, is not included in this study because neither the FRF archived spectra nor the existing model output (which stores 30 s-averaged sea-swell statistics) provide the time-series resolution required to separate the IG band. High-frequency instantaneous point output runs for selected events have been set up to enable IG spectral decomposition; results are deferred to a future study. The upper bound

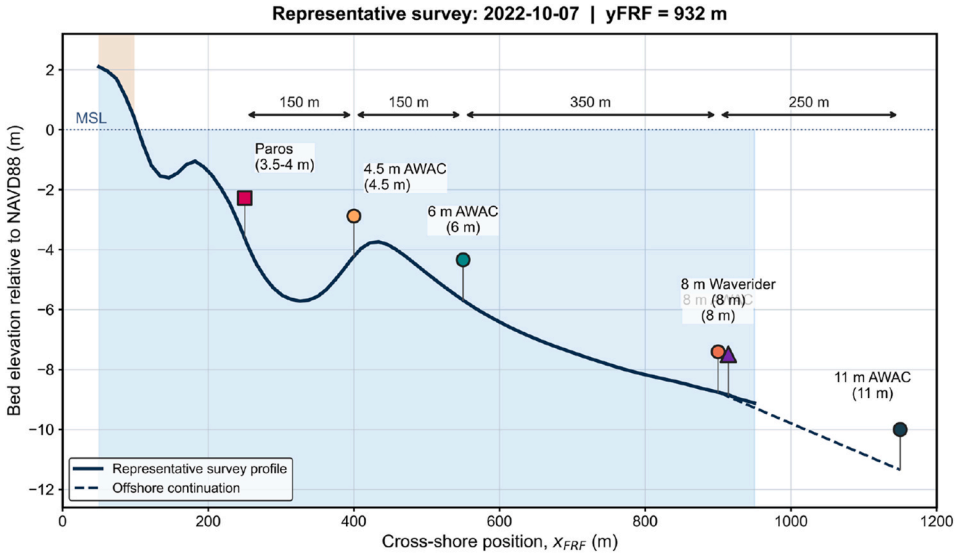


Fig. 2. Representative cross-shore bathymetric profile from the CRAB survey of 7 October 2022, the survey used by the largest number of selected events, at y_{FRF} = 932 m, the transect used for all simulations, showing the positions and nominal depths of all instruments used in this study. The shaded region indicates the active surf zone during moderate wave conditions. Dashed line indicates the offshore continuation of the profile beyond the survey extent. Distance annotations denote the inter-gauge segment lengths used in the energy flux gradient computation (Equation (4)). MSL: mean sea level (NAVD88 = 0 m).

of 0.33 Hz is set by the Nyquist frequency of the Paros pressure gauges, which sample at 2 Hz after anti-alias filtering; the AWAC profilers, which sample at higher rates, are spectrally truncated to this same upper limit to ensure consistency across all instruments in the array (Herbers et al., 2000).

The wave group velocity at each gauge position is computed from the linear dispersion relation using the peak period T_p and the local mean water depth h, corrected for the instantaneous tidal stage measured at the FRF pier tide gauge (Dean and Dalrymple, 1991). Using T_p rather than the zero-crossing or mean energy period systematically overestimates group velocity when the wave spectrum is broad or bimodal. Analysis of the 52-event record confirms that T_p-based C_g exceeds the T_{m02}-based value by a mean of 37 ± 23% across all gauge-event pairs (99.6% of records), with the largest overestimation at the 11 m AWAC (mean 52%) and the smallest at the Paros gauge (mean 19%). Since $K = H_{m0}/H_{m0}^{BC}$ involves no C_g term, the primary comparison metric is unaffected by this bias. The energy flux ratio Φ and the dissipation profiles $D(x)$, which both depend on EC_g, are affected. The bias does not cancel in the Φ ratio, even though the same T_p-based C_g is applied to the modeled and observed sides at each gauge. Cancellation would require the T_{m02}/T_p ratio to be constant across the array, whereas the spectra broaden shoreward, so the C_g overestimate varies with position rather than acting as a single multiplicative factor. Recomputing C_g from T_{m02} changes C_g by a median factor of 0.78 and does not leave Φ unchanged: the event-mean Φ rises from 0.87 to 1.04 at the 4.5 m AWAC, from 0.90 to 1.02 at the 6 m AWAC, and from 0.69 to 0.89 at the Paros gauge. On a T_{m02} basis the observed energy-flux retention is therefore close to unity until the inner surf zone, rather than declining progressively across the array as the T_p-based values suggest. Because the choice of period basis changes the qualitative shape of the profile and not merely its magnitude, Φ and $D(x)$ are not treated as quantitative skill metrics anywhere in this paper. They are retained only as qualitative descriptions of cross-shore structure, and no conclusion rests on them.

$$C_g = \frac{1}{2}\left(1 + \frac{2kh}{\sinh(2kh)}\right)\frac{\omega}{k} \tag{3}$$

where $\omega = 2\pi/T_p$ is the angular frequency and k is the wavenumber obtained by iterative solution of the dispersion relation $\omega^2 = gk\tanh(kh)$.

The discrete cross-shore dissipation rate between adjacent gauge pairs i and j, ordered from offshore to onshore, is then:

$$\overline{D}_{ij} = -\frac{\left[(EC_g)_j - (EC_g)_i\right]}{(x_j - x_i)} \tag{4}$$

where the overbar denotes spatial averaging over the inter-gauge segment. This yields a discrete profile of observed dissipation with up to six segments spanning the cross-shore array (Svendsen, 1984) (Fig. 2). All quantities are computed from 30-min burst-averaged spectra and then time-averaged over each selected event window to produce a single representative dissipation profile per event. A directional correction is applied to events where the mean wave direction deviates from shore-normal by an angle θ between 15° and 30°, following:

$$\overline{D}_{ij,corrected} = \frac{\overline{D}_{ij}}{\cos^2(\theta)} \tag{5}$$

This correction accounts for the oblique component of the energy flux and reduces the residual error in Equation (1) from up to 6% to less than 2% for events retained after the 30° directional threshold. Events with $\theta < 15°$ are used without correction. The $\cos^2\theta$ form does not follow from the energy balance. For waves obliquely incident on a coast with straight parallel contours, the quantity conserved in the cross-shore balance is the cross-shore component of the energy flux, $EC_g\cos\theta$, so Equation (1) becomes $d(EC_g\cos\theta)/dx = -D$. Estimating D from the raw flux difference over the cross-shore gauge separation, as Equation (4) does, omits that projection. Treating θ as constant over an inter-gauge segment gives $\overline{D}_{ij,\mathrm{corrected}} = \overline{D}_{ij}\cos\theta$, a single factor of $\cos\theta$ that reduces the estimate. Measuring the segment length along the wave ray rather than along the cross-shore axis introduces a compensating factor and leaves this result unchanged. Division by $\cos^2\theta$ acts in the opposite direction: at $\theta = 30°$ it increases the estimate by 33% where the derivation gives a 13% reduction. The $\cos^2\theta$ form was adopted empirically because it minimized the residual in Equation (1) for the retained events, and that is the only support it has. No conclusion in this paper is affected, because $D(x)$ and Φ are used only qualitatively (Section 2.3) and no skill statistic depends on them. A future quantitative use of these profiles should apply the single-cos θ projection and re-examine the residual. A single event-mean θ is used for all cross-shore segments; this approximation neglects depth-dependent refraction, which rotates the mean wave direction shoreward as depth decreases. The resulting systematic bias is opposite in sign at outer and inner segments (direction underestimated offshore, overestimated inshore) and is treated as a component of the observational uncertainty. This correction enters only the secondary dissipation and Φ metrics; the primary $K = H_{m0}/H_{m0}^{BC}$ comparison is a wave-height ratio that does not use it. Both its functional form, which projects the inter-gauge flux gradient onto the cross-shore direction for a single event-mean angle, and the neglect of depth-dependent refraction are acknowledged uncertainties confined to those secondary metrics, and the dissipation profiles are accordingly interpreted qualitatively.

2.2.1. Minimum segment requirement

An event is retained in the dissipation comparison only if a minimum of four out of six inter-gauge segments yield valid dissipation estimates after QC. When fewer than six segments are available, the skill score is computed over the available segments only, and the number of segments used is reported alongside each event result. Events with fewer than four valid segments are excluded from the dissipation comparison but retained in the Level 1 H_{m0} comparison if at least three gauge positions are available.

2.2.2. Observational uncertainty

The uncertainty in the gauge-derived dissipation estimate arises primarily from two sources: the spectral estimation of H_{m0} and the finite-difference approximation of the flux gradient. The relative uncertainty in H_{m0} is approximately 5% for the AWAC profilers and 3% for the Paros pressure gauges, based on manufacturer specifications and prior FRF instrument intercomparison studies. Propagating these uncertainties through Equations (2) and (4) using standard error propagation yields an uncertainty envelope on $\overline{D}_{ij}$ of approximately 10 to 15% for the inner surf zone segments, where the gradient is largest, and 5 to 10% for the outer segments. This envelope is included in all dissipation profile figures.

An additional source of uncertainty arises from the assumption of shore-normal propagation implicit in Equation (1). For events with mean wave direction within 30° of shore-normal, the cross-shore component of the energy flux gradient approximates the total gradient to within 6%, which is within the observational uncertainty. Events exceeding this directional threshold are excluded during event selection (Section 2.3).

2.2.3. Quality control

A three-step quality control procedure is applied before any dissipation profile is accepted. First, all 30-min records within the event window must carry a QC flag value of 1 (automated check passed) or 5 (human validated) in the FRF data files; records with flags of 3 (questionable) or 4 (failed) are rejected. Second, the offshore boundary H_{m0} at the 8 m Waverider buoy must exceed 0.3 m throughout the event window; records below this threshold correspond to negligible active

breaking in the surf zone. Third, the computed $\overline{D}_{ij}$ must be non-negative at a minimum of four of the six inter-gauge segments; negative dissipation in more than two consecutive segments is interpreted as contamination by tidal currents or measurement noise and the window is rejected. The fraction of rejected windows per event is reported alongside the results. This exclusion does not bias the retained sample toward monotonic profiles, because it never takes effect: across all 113,433 candidate windows the longest run of consecutive negative segments is two, so no window meets the rejection criterion of more than two and none is removed on this ground. Windows containing isolated negative segments are retained, 30,101 with a run of one and 1680 with a run of two, and these are exactly the non-monotonic profiles the criterion might have been expected to remove. The 24% of windows that are rejected are rejected for missing or low offshore H_{m0} (17,927 and 2040) or for an instrument QC flag of 3 or 4 (5875), none of which relates to the shape of the dissipation profile. Event selection does not use the dissipation quality control at all, so the criterion has no influence on which events entered the study.

2.3. Event selection

2.3.1. Eligibility criteria

Events are selected from the FRF record for the period January 2010 to December 2023. An event is defined as a continuous window of 3 to 6 h satisfying the following four criteria simultaneously: (1) Quasi-stationary forcing: the H_{m0} measured at the 8 m directional Waverider buoy varies by less than 15% within the window, computed as the coefficient of variation of 30-min burst values. (2) Near shore-normal incidence: the mean wave direction at the 8 m buoy deviates by less than 30° from the FRF shoreline normal (approximately 70° from true north). (3) Limited tidal stage change: the tidal water level measured at the FRF pier gauge changes by less than 0.3 m within the window, to minimize tidally driven depth changes between gauge positions. (4) Recent bathymetric survey: a CRAB bathymetric survey is available within 21 days of the event start time, to ensure the bathymetric input to the model represents the actual seafloor configuration during the event.

Application of these criteria to the full FRF record yields 86,281 eligible 3 to 6 h candidate windows distributed across the study period: 3893 energetic ($H_{m0} > 2.0$ m), 18,188 moderate ($1.0 \leq H_{m0} \leq 2.0$ m), and 64,174 quiescent ($0.3\ \text{m} \leq H_{m0} < 1.0$ m).

2.3.2. Sandbar state classification

A CRAB survey within 21 days is required for bathymetric input and morphological classification. At Duck, sandbars migrate at rates of order 10–50 m per week during energetic sequences (Gallagher et al., 1998; Plant et al., 1999). A 21-day window therefore introduces potential bathymetric positional errors of comparable magnitude to the bar migration distance; this positional uncertainty is unquantified and is acknowledged as a structural limitation. However, analysis of the 52-event record shows that the actual survey lags range from 0 to 14 days (mean 5.6 days). The Spearman rank correlation between survey lag and boundary transmission ratio K_{bnd} is $r_s = 0.15$ ($p = 0.30$, $n = 52$), so survey age within the window used here does not explain variance in model boundary performance. This null result does not rule out effects on cross-shore transformation skill for events with large morphological change between survey and event, but it supports the adequacy of the 21-day criterion for boundary energy comparisons.

Each candidate window is classified by sandbar morphological state using the closest CRAB survey within the 21-day window. The cross-shore bed elevation profile is analyzed to identify bar crests as local maxima exceeding the surrounding trough elevation by at least 0.4 m, within a cross-shore search window spanning $x_{FRF} = 50$ to 700 m (Plant et al., 1999). Events are classified as single-barred (one crest identified), double-barred (two crests identified), or unbarred (no crest meeting the relief threshold). Where the CRAB survey does not extend to $x_{FRF} = 700$ m, the search window is truncated at the seaward limit of the survey.

2.3.3. Stratified sampling design

Random sampling from the full candidate pool would yield a dataset dominated by quiescent events, which constitute 74% of all candidates, and would undersample energetic conditions. To ensure balanced representation across energy classes and sandbar states, a stratified random sampling design is applied. The candidate pool is partitioned into a 3×3 matrix of energy class by sandbar state, and events are drawn randomly from each cell with the target counts listed in Table 1. Within each cell, events are selected to maximize temporal spread across the full study period: candidates are ranked by year and sampled to ensure coverage across all available years rather than clustering in any single period. Additionally, within each energy class the selected events are required to span the full H_{m0} range of that class, with at least two events in the lower third and two events in the upper third of the class interval, to ensure that the skill score scatter plots in Section 3 have sufficient dynamic range (Cochran, 1977).

The double-barred cell target of 4 events per energy class is half the single-barred target and reflects the relatively rare occurrence of double-barred conditions in the FRF record. The design therefore samples double-barred conditions less densely than single-barred conditions by construction, and that imbalance is a property of the target matrix rather than a shortfall in filling it. The rule adopted in advance was that if fewer than 4 eligible candidates existed in any cell, all available candidates in that cell would be used and the shortfall reported in Section 3; in the event, no cell fell short.

2.3.4. Event inventory

The full list of selected events, including date, duration, offshore H_{m0}, T_p, mean direction, directional deviation from shore-normal, sandbar state, and the date of the CRAB survey used for model bathymetry, is provided in Supplementary Table S1 (full inventory including K_{bnd}). The distribution of events across the study period and across energy classes and sandbar states is summarized in Fig. 4.

2.4. XBeach non-hydrostatic mode

All simulations use XBeach in non-hydrostatic mode (*wavemodel = nonh*) in one horizontal dimension. The executable was built from the open-source Deltares.

2.4.1. Governing equations

XBeach-NH solves the depth-averaged nonlinear shallow water equations augmented with a non-hydrostatic pressure correction term (Smit et al., 2013). The governing momentum equation in the cross-shore direction is:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + g\frac{\partial \eta}{\partial x} + \frac{1}{\rho}\frac{\partial q}{\partial x} = \frac{\tau_b}{\rho h} \tag{6}$$

where u is the depth-averaged cross-shore velocity (m s^{-1}), η is the free surface elevation (m), q is the depth-averaged non-hydrostatic pressure (Pa), τ_b is the bed shear stress (Pa), and $h = \eta + d$ is the total water depth

Table 1
Target event counts per stratified cell. Every cell target was met exactly, so the achieved counts are identical to the targets shown and no cell was short (Section 3.3.2).

Energy class	Hm0 range (m)	Single-barred	Double-barred	Unbarred	Total
Energetic	> 2.0	8	4	4	16
Moderate	1.0–2.0	8	4	4	16
Quiescent	0.3–1.0	8	4	8	20
Total		**24**	**12**	**16**	**52**

with d the still-water depth. The continuity equation is:

$$\frac{\partial \eta}{\partial t} + \frac{\partial (hu)}{\partial x} = 0 \tag{7}$$

Wave breaking is represented implicitly through the momentum-conservative numerical scheme, which allows wave fronts to steepen into bores when the local wave steepness exceeds the prescribed threshold $\beta_{\max}$. Unlike phase-averaged models, no separate analytical dissipation formula is required; energy loss by breaking emerges from the numerical solution of Equations (6) and (7) once the steepness criterion is met. This is the feature that makes XBeach-NH suitable for phase-scale dissipation assessment: the model dissipation is a direct output of the resolved wave dynamics rather than a parameterized source term.

2.4.2. Model grid and bathymetry

A one-dimensional cross-shore model grid is constructed for each event using the CRAB survey DEM closest in time within the 21-day window. The primary comparison transect is the single survey row nearest $y_{FRF} = 940$ m, which on the 4 m survey grid is $y_{FRF} = 932$ m; bed elevations along that row are linearly interpolated onto the model grid at the target resolution. No alongshore averaging is applied, so the modeled bathymetry is the surveyed profile rather than a smoothed composite. This transect is co-located with the permanent cross-shore gauge array, whose alongshore position is $y_{FRF} \approx 940$ m, and is the most representative of open-ocean conditions away from the FRF pier (located at $y_{FRF} \approx 530$ m), which is sufficiently distant for the transect to be free of its hydrodynamic influence. All simulations reported in this study use this transect. The alongshore bathymetric variability at the site is characterized separately in Section 3.3.3 from nine survey transects spanning the DEM domain, which provide morphological context for the primary transect but are not simulated.

The grid resolution varies with distance from the shoreline: 1 m over the 300 m closest to the shoreline, 5 m from 300 to 600 m, and 10 m seaward of 600 m. For a representative event this places the 1 m zone at $x_{FRF} = 104$ to 404 m, the 5 m zone at 404 to 704 m, and the 10 m zone at 704 to 950 m, yielding 385 grid cells on average (range 346 to 388 across the 52 events). This resolution is sufficient to resolve individual sandbar features, whose typical cross-shore widths exceed 20 to 30 m at the FRF. The offshore boundary is placed at the seaward limit of the survey coverage, $x_{FRF} = 950$ m, where the bed lies at − 9.07 m NAVD88 on average (range −9.30 to −6.70 m) and the still-water depth including the event mean water level is 9.2 m on average (range 6.8 to 10.2 m). The boundary is therefore 50 m seaward of the 8 m AWAC ($x_{FRF} = 900$ m), the most offshore gauge inside the model domain, which provides at least 50 m of propagation between the boundary and the first comparison gauge. The 11 m AWAC ($x_{FRF} = 1150$ m) lies outside the domain and is used as an observational offshore reference only.

2.4.3. Boundary conditions

The offshore boundary condition is a time series of free surface elevation reconstructed from the directional variance density spectrum measured at the 8 m Waverider buoy. The spectrum is transformed to the boundary depth using linear shoaling over the intervening bathymetry, then a random-phase realization of duration equal to the event window plus a 30-min spinup period is generated by:

$$\eta(t) = \sum_n a_n \cos(2\pi f_n t + \varphi_n) \tag{8}$$

where $a_n = \sqrt{2S(f_n)\Delta f}$ is the amplitude of the n th frequency component, f_n is the discrete frequency, and φ_n is a random phase drawn from a uniform distribution on $[0,\ 2\pi]$. The resulting signal is verified to reproduce the target H_{m0} and T_p to within 2% before each model run. A time-varying mean water level offset derived from the pier tide gauge record is applied uniformly across the model domain to account for tidal stage during each event.

The mean angle of wave incidence θ_m at the offshore boundary (*mainang* in the XBeach parameter file) is set for each event to the mean compass bearing measured at the 8 m Waverider buoy, rotated into the model coordinate system such that shore-normal incidence corresponds to $\theta_m = 0°$. The phase realization is not controlled. No random keyword is set in any of the parameter files, so the model default of random = 1 applies and the seed for the phase draws is taken from the system clock at run time. Prescribing the boundary through a jonstable table fixes the bulk parameters, H_{m0}, T_p, mean direction, peak enhancement, and spreading, but not the phases, which the model reconstructs internally. The boundary time series is therefore not bitwise reproducible between runs; the consequences for the reported scatter are assessed in Section 3.2.

2.4.4. Model parameters

All model parameters are set to physically motivated values from literature and held constant across all events. No per-event calibration is performed. The key parameters and their justifications are listed in Table 2.

The 1-layer non-hydrostatic formulation is used throughout. The 2-layer extension of de Ridder et al. (2021), which improves dispersion in intermediate water, is a documented alternative for events where kh at the offshore boundary exceeds 1.0, the nominal limit of the 1-layer formulation. It is not used for the production runs, but the two formulations are compared directly in the flat-bottom diagnostic of Section 2.10 to test whether the single-layer dispersion limit contributes to the boundary deficit.

2.5. Numerical sensitivity and convergence tests

2.5.1. Grid resolution

The standard model grid uses the variable resolution defined in Section 2.4.2, 1 m over the 300 m closest to the shoreline, 5 m from 300 to 600 m, and 10 m out to the offshore boundary at $x_{FRF} = 950$ m, yielding 385 cells on average. Two alternative grids were run for three

Table 2
XBeach model parameters used in this study. The third column indicates whether the parameter acts on the solution when wavemodel = nonh. The three parameters marked "No" belong to the surfbeat formulation and are listed because they are used by the hydrostatic ablation of Section 2.8, not because they influence the non-hydrostatic runs.

Parameter	Value	Active in NH mode	Basis
Breaking steepness threshold ($\beta_{\max}$)	0.6	Yes	Default value in Smit et al. (2013)
Manning bed friction coefficient (n)	0.015 m − 1/3 s	Yes	Representative sandy beach value after Ruessink et al. (2001)
Nonhydrostatic pressure formulation	Single layer	Yes	Following Smit et al. (2013); two-layer alternative of de Ridder et al. (2021) tested in Section 3.12 only
Offshore boundary condition	Random phase	Yes	Defined according to Eq. (8)
Spin up duration	30 min	Yes	Standard XBeach practice
Lateral boundary condition	1D wall	Yes	Consistent with the shore-normal forcing criterion used for event selection
Breaker index (γ)	0.55	No	Surfbeat wave-action balance only; used in the hydrostatic ablation of Section 2.8
Roller dissipation coefficient	1.0	No	Surfbeat roller model only; used in the hydrostatic ablation of Section 2.8
Maximum wave height to depth ratio $\gamma_{\max}$	2.0	No	Surfbeat wave-action balance only; used in the hydrostatic ablation of Section 2.8

representative events: event_0010 (energetic, H_{m0} = 4.25 m, single-barred), event_0038 (moderate, H_{m0} = 2.00 m, double-barred), and event_0050 (quiescent, H_{m0} = 0.60 m, unbarred). The coarse grid uses $\Delta x_{\min}$ = 2 m and $\Delta x_{\max}$ = 20 m ($n_x \approx 193$ cells); the fine grid uses $\Delta x_{\min}$ = 0.5 m and $\Delta x_{\max}$ = 5 m ($n_x \approx 770$ cells). All other parameters, bathymetry, and boundary conditions are identical to the standard run.

For the coarse grid, H_{m0} profile RMSE relative to the standard grid was 0.259 m for event_0010, 0.502 m for event_0038, and 0.121 m for event_0050; corresponding mean absolute differences were 0.201 m, 0.497 m, and 0.118 m. The comparatively large coarse-grid deviations for event_0038 reflect that the double-barred bathymetry imposes short spatial scales on the breaking region that the 20 m offshore cells do not resolve adequately. The fine grid did not complete for event_0010: the run terminated at t = 1250 s of the 10800 s simulation when the time step collapsed to order 10^{-6} s following a velocity blow-up near x_{FRF} = 471 m, so no fine-grid profile is available for the energetic case. For event_0038 the fine grid produced an RMSE of 0.310 m and a mean absolute difference of 0.298 m relative to the standard grid, and for event_0050 the corresponding values were 0.237 m and 0.190 m. The fine-grid deviations for event_0050 are slightly larger than the coarse-grid deviations, consistent with spatial averaging of the low-amplitude signal over the finer output grid rather than with a convergence failure. Across the two events for which both alternative grids are available, the standard grid sits within the envelope bracketed by the coarse and fine grids and produces cross-shore K profiles that are structurally identical to the fine-grid solution. These results confirm that the standard grid is adequate for the resolution of cross-shore wave transformation at this site.

2.5.2. Spin-up duration

The standard model spin-up is 30 min (t_{start} = 1800 s), during which the boundary signal is applied but model output is suppressed. Spin-up durations of 15 min and 60 min were tested for the same three representative events to determine whether 30 min is sufficient for the wave field to reach a statistically steady state across the profile.

The mean absolute H_{m0} difference between the 15-min and 30-min spin-up runs was 0.0220 m for event_0010, 0.0164 m for event_0038, and 0.00489 m for event_0050. The corresponding values for the 60-min versus 30-min comparison were 0.0346 m, 0.0184 m, and 0.00108 m. For all three events the differences are small relative to the observational uncertainty in H_{m0} (approximately 5% of the local wave height), and no systematic cross-shore structure is present in the difference profiles. The 30-min spin-up is therefore adequate for the sea-swell energy levels and profile lengths encountered in this study.

2.5.3. Parameter sensitivity

Two model parameters act on the solution in XBeach non-hydrostatic mode (*wavemodel* = *nonh*) and are therefore the subject of the sensitivity analysis. Wave breaking onset is controlled by the breaking steepness threshold $\beta_{\max}$ (*maxbrsteep*), which determines when a wave front steepens into a bore and is the only tunable breaking parameter in this mode, and bed dissipation is controlled by the Manning coefficient n (*bedfriccoef*). The parameters of the surf-beat formulation, the breaker index γ (*gamma*), the roller dissipation coefficient, and the maximum wave-height-to-depth ratio $\gamma_{\max}$ (*gammax*), act on the wave-action balance and the roller model and have no effect when the non-hydrostatic solver is selected. They are used only in the hydrostatic surfbeat ablation of Section 2.8, where they are reported with the values adopted there, and they are not perturbed in NH mode because they cannot influence the NH solution. This distinction is stated here because these parameters appear in both configurations and only one of the two uses them.

One-at-a-time perturbations were therefore applied to $\beta_{\max}$ and n only. Six events spanning the three energy classes were perturbed, two per class: event_0011 (H_{m0} = 4.56 m) and event_0031 (2.85 m) in the energetic class, event_0038 (2.00 m) and event_0044 (1.34 m) in the moderate class, and event_0043 (1.00 m) and event_0050 (0.60 m) in the quiescent class. For each event, $\beta_{\max}$ was set to 0.4 and 0.8, bracketing the standard value of 0.6, and n to 0.010 and 0.025 m $-1/3$ s, bracketing the standard 0.015 m $-1/3$ s, giving 24 perturbation runs. Each perturbation used the same jonstable boundary specification as the corresponding baseline run, with all other settings held constant. The phase realization is not fixed between runs (Section 2.4.3), so each member of a perturbation pair differs from its baseline in the phase draw as well as in the perturbed parameter. The responses reported in Section 3.7 therefore include a phase-realization component and should be read as upper bounds on the parameter effect. The results are reported in Section 3.7.

2.5.4. Forcing type comparison

The standard boundary condition is a time-varying random-phase JONSWAP realization constructed from the event-period wave spectrum following Equation (8). To assess whether the time-varying implementation introduces systematic differences relative to a simpler stationary boundary, a single-spectrum stationary run was executed for event_0010 and event_0038 using a monochromatic JONSWAP spectrum matched to the event-mean H_{m0} and T_p.

For event_0010, the mean absolute H_{m0} difference between the stationary and transient runs was 0.0387 m (RMSE = 0.124 m) across 387 co-located profile points. For event_0038 the mean absolute difference was 0.0295 m (RMSE = 0.0303 m) across 386 profile points. The larger absolute differences for event_0010 reflect the higher energy level rather than a qualitative disagreement in profile shape. When the profiles are normalized to produce $K = H_{m0}/H_{m0}^{BC}$, the mean absolute K difference is 0.0185 for event_0010 and 0.00730 for event_0038, confirming that the cross-shore transformation structure is preserved under both forcing implementations. The stationary and transient runs produce K profiles that are in close agreement from the offshore boundary through the surf zone, and the boundary-deficit behavior documented throughout this study is present in both.

2.6. Comparison metrics

As established in Section 3.2, the direction-dependent energy transmission deficit at the 1D JONSWAP boundary precludes absolute H_{m0} and dissipation comparisons. This deficit was discovered post-run; the normalized metrics below were adopted in response to it. Accordingly, all skill assessments using these metrics are treated as exploratory diagnostic findings and are not presented as pre-specified confirmatory tests.

Model skill is evaluated on three levels, which answer different questions and must not be conflated. The distinction matters because the boundary deficit enters the modeled and observed sides of the comparison asymmetrically. The first level does not isolate transformation error.

The first level is the boundary-referenced ratio:

$$K = \frac{H_{m0}}{H_{m0}^{BC}} \tag{9}$$

where H_{m0}^{BC} is the significant wave height prescribed at the offshore boundary for each event. K is computed at each gauge position for both model and observations using the same H_{m0}^{BC} denominator. $K = 1$ represents no change relative to the prescribed boundary height; $K > 1$ indicates shoaling amplification; $K < 1$ indicates net dissipation. The observed K is referenced to a quantity the waves actually carried, because the offshore boundary height is taken from the measured offshore spectrum, so observed K starts near unity at the seaward end of the array. The modeled K is not: the wave-generation boundary delivers only a fraction K_{bnd} of the prescribed height (Section 3.2, sample mean 0.57), so the modeled profile starts at K_{bnd} rather than at 1. Dividing both sides by the same prescribed H_{m0}^{BC} therefore does not remove the

boundary deficit from the comparison; it leaves the deficit intact as an offset between two profiles that begin at different values. K is consequently a diagnostic of the combined boundary-delivery and cross-shore-transformation error, and is reported as such. The mean modeled K bias of −0.48 to −0.64 (Section 3.4) is of the same size as $1 - K_{bnd}$, so most of that bias is boundary under-delivery rather than transformation error. This changes the meaning of the near-zero skill reported in Section 3.4. It is a null result about the combined quantity, not evidence that the cross-shore physics is wrong.

The second level is the delivered-energy-referenced transformation ratio, which is the metric this study uses to assess cross-shore transformation proper:

$$K_{res} = \frac{K_{model}}{K_{bnd}} \tag{10}$$

where K_{bnd} is the event-specific boundary transmission ratio defined in Section 3.2. Referencing the modeled profile to the energy the boundary actually delivered, rather than to the energy requested, puts the modeled and observed profiles on the same footing. Both then start near unity at the seaward end and differ only through what happens across the surf zone. K_{res} is the primary transformation metric of this study and carries its own bootstrap statistics and its own scatter comparison. Two limitations are stated once and carried throughout. First, because K_{bnd} is computed from the same run, the reduction in mean bias that accompanies the rescaling is arithmetic and is not evidence of skill; only the event-scale correlation with the observations is informative. Second, rescaling by an event-mean constant cannot undo the nonlinear consequences of having propagated deficient energy through breaking, setup, and infragravity generation, which are threshold-dependent.

The third level is the corrected-boundary experiment of Section 3.10, in which the prescribed boundary height is inflated by $1/K_{bnd}$ and the full event set is re-run so that the intended energy is delivered to the same solver. This is an a posteriori sensitivity experiment rather than a validation, because the inflation factor is derived from the runs being corrected. Section 3.11 supplies the closest available approximation to an independent test by predicting K_{bnd} from the prescribed forcing direction alone and scoring the correction only on withheld events.

A normalized energy flux ratio $\Phi = EC_g/EC_{g,\mathrm{offshore}}$ is also computed, where the offshore reference is the mean EC_g at $x_{FRF} \geq 940$ m from model output and the measured EC_g at the 11 m AWAC ($x_{FRF} = 1150$ m) from observations. For the reasons given in Section 2.3, Φ and the dissipation profiles depend on the choice of period basis strongly enough that their cross-shore shape changes with it. They are therefore used only qualitatively and carry no skill statistics. Φ is computed for energetic and moderate events only; 11 of the 20 quiescent events have an offshore reference flux below 200 W/m, where the near-zero denominator makes the ratio unstable.

For the multi-event skill assessment, bias = mean (modeled − observed), RMSE, and Pearson correlation coefficient r are computed across all gauge-event pairs within each energy class. Gauge-event pairs are not independent observations. Several gauges are compared within a single event, and they share one boundary realization, so an inference framework that treats the pairs as independent overstates the effective sample size. All uncertainty reported in this paper therefore comes from a single framework, an event-block bootstrap that resamples the 52 events with replacement (20,000 iterations) and recomputes the statistic on the resampled event blocks. The framework is applied identically to r, to bias, and to RMSE, and to every pair set, so that no result depends on a pairwise-independence assumption. Two-sided cluster-robust p-values are obtained from the bootstrap distribution as twice the smaller tail fraction about zero. Fisher z-transformed intervals and t-tests on $n - 2$ degrees of freedom are not reported: for these data they are narrower than the event-block intervals and would attribute significance to correlations the clustered design cannot resolve.

Two gauge-event pair sets are used, and because they carry different sample sizes both are defined here once and identified by name wherever they appear. The in-domain pair set supports the model-data transformation comparison of Section 3.4. It requires that a gauge lie inside the model domain and that both an observed and a modeled K exist for that gauge and event. There are 260 possible combinations, 52 events at five gauge positions. Of these, 52 are removed because the 11 m AWAC lies seaward of the offshore boundary and is therefore never modeled, 98 because no valid observed H_{m0} is available at that gauge during that event, and 2 because the gauge falls outside the model grid for that particular event. That leaves $n = 110$ pairs (36 energetic, 34 moderate, 40 quiescent). The full valid-observation pair set supports the hydrostatic-baseline and corrected-boundary comparisons of Sections 3.8 and 3.10, which are model-versus-model contrasts scored against the same observations. It retains every gauge-event pair with a valid observed K, including the 45 pairs at the 11 m AWAC, giving $n = 203$ pairs (65 energetic, 63 moderate, 75 quiescent) over the same 52 events. The larger set is used there so that both models are scored on an identical set of observations. The near-zero uncorrected skill is reproduced on both sets, so no conclusion depends on the choice: the pooled correlation is 0.08 (95% CI −0.05 to 0.20) on the in-domain set and 0.04 (−0.06 to 0.13) on the full set. Every table and figure caption states which set it uses.

2.7. Hydrostatic baseline model

To test whether the near-zero cross-shore transformation skill reported in Section 3.4 reflects a limitation specific to the 1D non-hydrostatic configuration or a more general shortcoming of the normalized assessment, a phase-averaged hydrostatic baseline was computed for all 52 events on the same CRAB bathymetry and from the same prescribed boundary wave height. The two models are not forced in the same sense, and the distinction is central to how the comparison must be read: the baseline is initialized at the offshore node with the full prescribed H_{m0}^{BC}, whereas XBeach-NH is driven by a wave-generation boundary that delivers only a fraction of it (Section 3.2). The baseline therefore starts from the intended offshore energy by construction and the non-hydrostatic model does not. The baseline solves the stationary cross-shore energy balance of Equation (1) from the offshore boundary to the shoreline, with the dissipation term given by the Battjes and Janssen (1978) random-wave bore model:

$$D = \frac{\alpha}{4} f_p \, \rho g \, Q_b \, H_m^2 \tag{11}$$

where $f_p = 1/T_p$ is the peak frequency, $\alpha = 1$ is the dissipation coefficient, $H_m = \gamma h$ is the depth-limited maximum wave height, and Q_b is the fraction of breaking waves obtained from the implicit relation:

$$\frac{1 - Q_b}{\ln Q_b} = -\left(\frac{H_{rms}}{H_m}\right)^2 \tag{12}$$

Group velocity and shoaling are evaluated from linear dispersion using T_p and the local depth, consistent with the observational processing in Section 2.2. The balance is integrated with two breaker indices that bracket common field values, $\gamma = 0.42$ (Battjes and Janssen, 1978; $\gamma = 0.50$Thornton and Guza, 1983$\gamma = 0.50$), and is initialized with the prescribed boundary wave height, $H_{rms} = H_{m0}^{BC}/\sqrt{2}$, at the offshore node. Because the baseline is forced directly with the prescribed boundary energy, it is by construction not subject to the 1D non-hydrostatic boundary transmission deficit documented in Section 3.2. This is intentional: it isolates the cross-shore shoaling and breaking response from the boundary effect. The modeled $K = H_{m0}/H_{m0}^{BC}$ profile is interpolated to the gauge positions and scored against observations using the same bias, RMSE, and Pearson r statistics applied to XBeach-NH (Section 2.6).

2.8. In-model hydrostatic ablation

The parametric baseline of Section 2.7 is initialized at the offshore node with the full prescribed boundary energy, so it isolates the cross-shore breaking and shoaling response but cannot test whether the boundary transmission deficit of Section 3.2 is specific to the non-hydrostatic solver. To separate the boundary deficit from the wave-resolving dynamics, a complementary in-model ablation was run for the three representative events of Section 3.1 using XBeach in surfbeat mode (hydrostatic short-wave envelope with the roller model), on the identical grid, bathymetry, tidal level, and jonstable boundary table used for the non-hydrostatic runs. The non-hydrostatic pressure solve is gated only on the wave-model selector, so surfbeat provides a true hydrostatic propagation of the same forcing rather than a perturbation of the non-hydrostatic configuration. Wave breaking used the roller dissipation formulation as implemented in XBeach (Roelvink et al., 2009) with $\gamma = 0.55$, $\alpha = 1.0$, $n = 10$, and $\gamma_{\max} = 2.0$, on a directional grid spanning $\pm 90^{\circ}$ at 15° resolution. The remaining settings were absorbing-generating boundaries at both ends, Manning friction $n = 0.015$ m $-1/3$ s, morphology disabled, a 30-min spin-up, and CFL = 0.7, matching the non-hydrostatic configuration wherever the two modes share a parameter. In surfbeat mode the model short-wave variable is the root-mean-square height, so the significant wave height was recovered as $H_{m0} = \sqrt{2}\, H_{rms}$, distinct from the phase-resolved estimate $H_{m0} = 4\sqrt{\zeta'^2}$ used in non-hydrostatic mode (Section 2.6). The boundary transmission ratio K_{bnd} was computed using the canonical definition of Section 3.2, the mean over the three most offshore nodes relative to the prescribed boundary H_{m0}.

2.9. Corrected-boundary comparison

Neither the parametric baseline of Section 2.7 nor the in-model ablation of Section 2.8 supplies the correct offshore energy to the non-hydrostatic solver itself: the baseline runs a hydrostatic solver, and the ablation propagates the same deficient boundary through both modes. To provide the controlled test, the full 52-event set was re-run in a corrected-boundary configuration. For each event the prescribed boundary H_{m0} in the jonstable table was multiplied by the per-event factor $1/K_{bnd}$, where K_{bnd} is the measured boundary transmission ratio of Section 3.2 (Supplementary Table S1), so that the energy delivered to the offshore nodes matches the originally intended target. Only the boundary H_{m0} was altered; the period, mean direction, directional spreading, peak-enhancement factor, bathymetry, tidal level, wind forcing, grid, and all numerical parameters were copied unchanged from the corresponding baseline run. The corrected runs were post-processed identically to the baseline runs (Section 2.6) and scored against the same observed K and gauge-event pairs (Section 2.6, Section 3.8). Events for which the inflated forcing collapsed at the offshore boundary, identified by a first-node H_{m0} exceeding the near-offshore plateau (the median of nodes 4 to 9) by more than a factor of 1.5, were flagged as invalid corrections and excluded from the corrected skill statistics; results are reported both with and without these events.

2.10. Flat-bottom boundary diagnostic

To determine whether the boundary transmission deficit of Section 3.2 originates in the cross-shore bathymetry and breaking physics or in the 1D wave-generation boundary itself, a controlled flat-bottom experiment was run in the identical solver configuration (*wavemodel* = nonh, *wbctype* = jonstable, *nonhspectrum* = 1, ARC = 1, *ny* = 0, *maxbrsteep* = 0.6, Manning $n = 0.015$). The cross-shore bathymetry was replaced by a horizontal shelf at 12 m depth backed by a dissipative beach, which removes depth-induced shoaling and breaking at the offshore nodes. The forcing was held at a single low sea state ($H_{m0}^{BC} = 1.0$ m, $T_p = 10$ s, $\gamma_{jsp} = 3.3$, $s = 10$) so that no depth-limited breaking occurs at the boundary. The only quantity varied is the mean wave angle θ_m (*mainang*), swept from -30° to $+30^{\circ}$ in 5° steps, with K_{bnd} computed using the canonical three-node definition. A second sweep varied the directional spreading s from 10 to 1000 at $\theta_m = 0^{\circ}$ to isolate the contribution of directional content. A further set of shore-normal controls varied the boundary time step (*dtbc* from 0.25 to 1.0 s), the wave-generation order, the boundary-condition version, spectral versus non-spectral generation, and the spectral width, to test whether the baseline deficit is a removable configuration choice. Because the domain contains no sandbars, refraction, or storm-track covariation, any K_{bnd} structure produced is attributable to the wave-generation boundary alone.

Two further sweeps on the same flat-bottom configuration test explanations that the angle sweep alone cannot exclude. The first varies the depth of the horizontal shelf over 9, 12, 17, and 25 m with the forcing, solver settings, and dissipative beach held fixed, at $\theta_m = 0^{\circ}$ and $\theta_m = +20^{\circ}$, to test whether the deficit is a consequence of placing the offshore boundary too close to the breaking zone. If the deficit arises because the prescribed sea state is marginal at the boundary depth, moving the boundary into deeper water must reduce it. The second varies the non-hydrostatic layer count, comparing the single-layer formulation used throughout against the reduced two-layer formulation of de Ridder et al. (2021), available in the same build as *nonhq3d* = 1 with *nhlay* at its default. This isolates the single-layer dispersion limit from the generation artifact, because the two hypotheses make opposite predictions: a dispersion limitation must grow with kh and shrink under the two-layer formulation, whereas a generation artifact does neither. The comparison is run at shore-normal incidence across peak periods of 4, 5, 6, 8, 10, and 14 s at $H_{m0}^{BC} = 1.0$ m, spanning $kh = 0.52$ to 3.03 at the 12 m shelf, plus an energetic case at $H_{m0}^{BC} = 4.0$ m and $T_p = 12$ s. Each configuration is run with three independent phase realizations, and the spread across realizations is reported so that differences between configurations can be judged against it. In this sweep the two nodes inshore of the generation node carry large transients in both the single-layer and the two-layer configuration, which makes the three-node average unstable, so the layer-count comparison alone is scored at the wave-generation node rather than by the canonical three-node definition. The two conventions are close where both are well behaved: across the depth sweep they differ by at most 0.013. Because the layer-count comparison is a difference between two configurations scored the same way, the choice of node convention does not affect its conclusion.

2.11. Data availability and reproducibility

All FRF observational data used in this study are publicly available without restriction via the USACE Coastal and Hydraulics Laboratory THREDDS server. XBeach-NH is open-source and freely available at https://xbeach.readthedocs.io. The complete 52-event inventory, including all metadata columns and the computed K_{bnd} for each event, is provided as Supplementary Table S1.

3. Results

3.1. Representative event profiles

Fig. 3 shows modeled H_{m0} profiles against the cross-shore bathymetry for three events that bracket the range of forcing conditions in the 52-event set: energetic event_0010 ($H_{m0}^{BC} \approx 4.3$ m), moderate event_0038 ($H_{m0}^{BC} \approx 2.0$ m), and quiescent event_0050 ($H_{m0}^{BC} \approx 0.6$ m). All three profiles are from the baseline standard-grid run. These events are selected to illustrate how the model responds across genuinely distinct hydrodynamic regimes rather than to identify best or worst cases. Event_0010 shows near-boundary wave height collapse consistent with individual wave heights exceeding the depth-limited breaking threshold

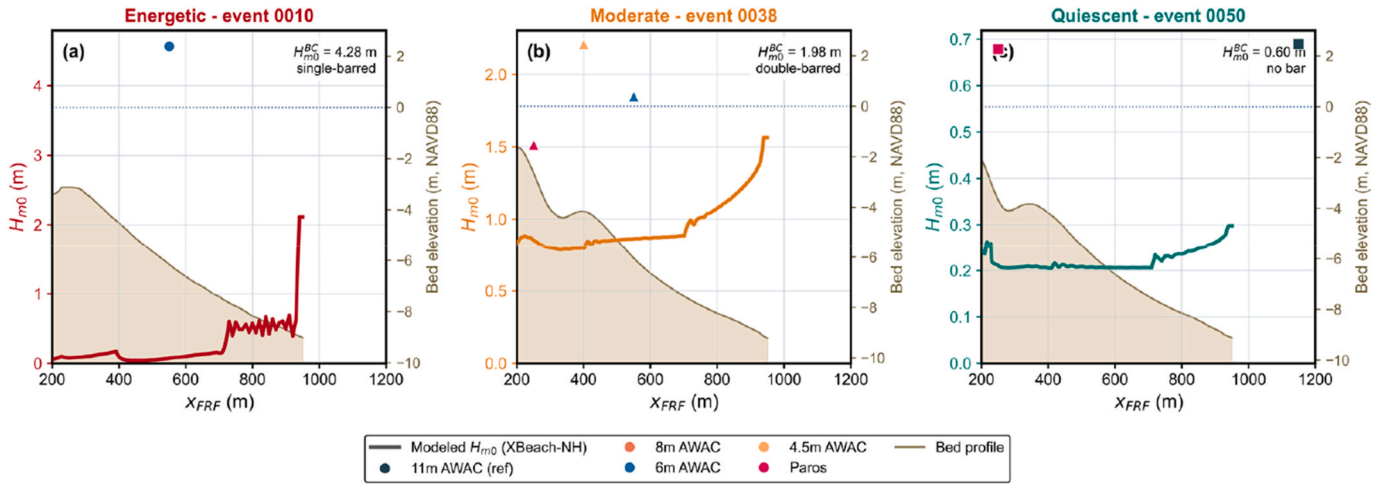


Fig. 3. Modeled cross-shore significant wave height (H_{m0}) profiles for the three representative events that bracket the forcing range: (a) energetic event_0010 ($H^{BC}_{m0} = 4.3$ m, single-barred), (b) moderate event_0038 ($H^{BC}_{m0} = 2.0$ m, double-barred), and (c) quiescent event_0050 ($H^{BC}_{m0} = 0.6$ m, unbarred). Each panel shows the modeled H_{m0} (colored line, left axis) over the underlying CRAB bed profile (shaded, right axis), with observed gauge H_{m0} overlaid as symbols. The energetic event exhibits near-boundary wave height collapse, whereas the moderate and quiescent events show nearly flat cross-shore profiles.

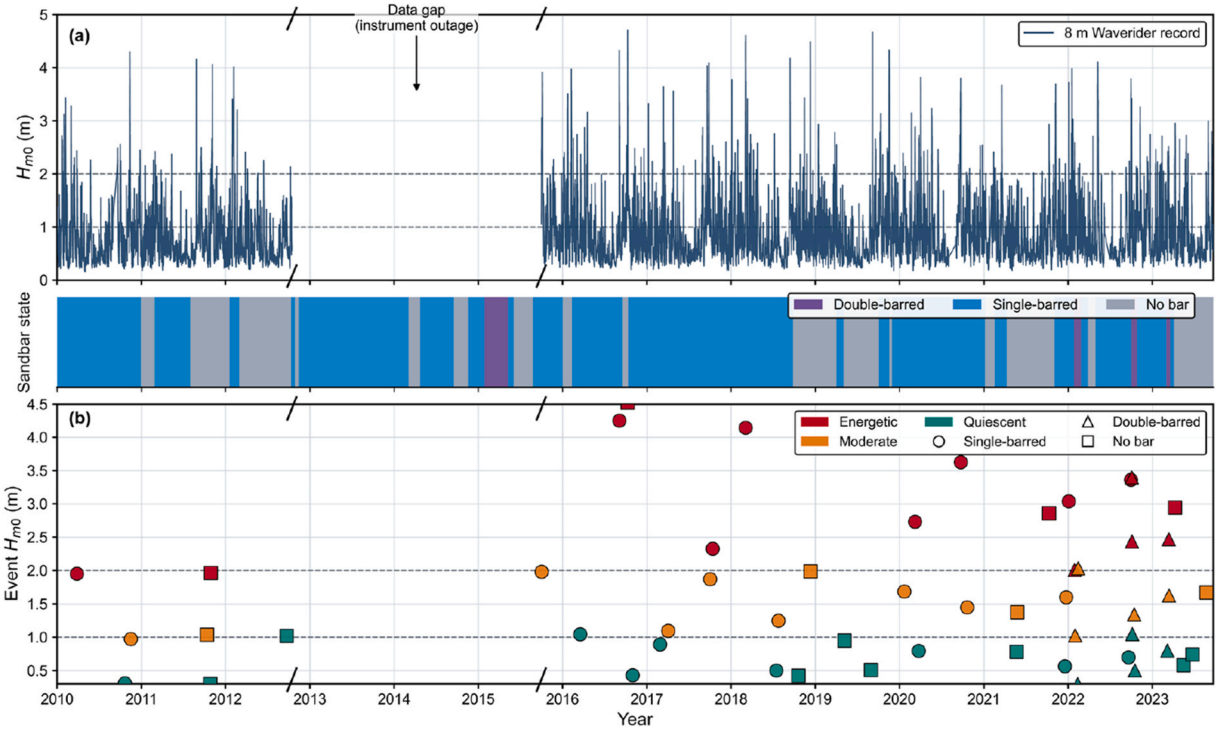


Fig. 4. Overview of the 8 m Waverider buoy record at the FRF Duck site, 2010 to 2023. Panels show (a) significant wave height Hm0 with dashed lines at the energy class thresholds of 1.0 m and 2.0 m, (b) peak period T_p, and (c) mean wave direction with dashed line indicating the FRF shoreline normal at 70°. Colored vertical markers indicate selected event times, classified as energetic (red, $H_{m0} > 2.0$ m), moderate (orange, $1.0 \leq H_{m0} \leq 2.0$ m), or quiescent (teal, $H_{m0} < 1.0$ m). The lower strip shows the sandbar morphological state of the closest CRAB survey at each event: blue (single-barred), purple (double-barred), grey (unbarred). The data gap from mid-2013 to early 2016 reflects an instrument outage. Direction values exceeding 200° are omitted as they correspond to data quality issues or anomalous storm passages.

$H/h \approx 0.6$ in approximately 10 m water depth. Event_0038 and event_0050 show the more typical pattern in which the modeled profile is nearly flat cross-shore. Together these three cases span the full range of boundary transmission ratios, breaking behavior, and morphological states present in the event set and serve as the reference cases for the grid-resolution and spin-up tests reported in Section 3.6.

3.2. Boundary condition transmission deficit

Post-run analysis of the 52-event baseline revealed a systematic wave-height transmission deficit at the offshore model boundary. XBeach-NH in non-hydrostatic mode with a JONSWAP table boundary condition generated, on average, approximately 57% of the prescribed offshore H_{m0} at the model boundary ($x_{FRF} = 950$ m). The boundary transmission ratio, defined as:

$$K_{bnd} = \frac{H_{m0,\text{model boundary}}}{H_{m0}} \quad (13)$$

ranged from 0.38 to 0.76 across the 52 events. Here $H_{m0,\text{model boundary}}$ is the significant wave height produced by the model, computed as the mean over the three most offshore grid nodes after simulation. These are active computational nodes, not ghost or boundary-storage points: the first is the wave-generation node itself at $x_{FRF} = 950$ m and the next two are the adjacent interior nodes at $x_{FRF} = 940$ and 930 m, so the average spans the 20 m immediately shoreward of the boundary. This three-node definition is canonical and is applied to every K_{bnd} reported in this study, including Supplementary Table S1, the in-model ablation of Section 3.9, the corrected-boundary runs of Section 3.10, and the flat-bottom sweep of Section 3.12, so all reported K_{bnd} values are directly comparable. The one exception is the layer-count comparison of Section 3.12, which is scored at the wave-generation node alone because the adjacent interior nodes are unstable in that configuration, for the reasons given in Section 2.10.

Investigation of the event-by-event variability in K_{bnd} revealed a strong dependence on the signed wave approach direction relative to shore-normal, and specifically on the sign of the directional offset rather than its magnitude. After controlling for offshore H_{m0} magnitude, the partial correlation between compass bearing and K_{bnd} was $r = -0.937$. The dependence is on the signed offset. K_{bnd} correlates strongly with the signed deviation from shore-normal ($r = -0.86$ across all 52 events, and $r = -0.93$ with the two near-boundary-collapse events identified below excluded), but is effectively uncorrelated with the absolute deviation ($r = 0.09$). This is consistent with the absence of a significant difference between absolute-deviation bins (one-way ANOVA $p = 0.45$; Fig. S4). Grouped by sign, events with a negative offset ($\theta_m < -5°$) transmitted a mean K_{bnd} of 0.63 ($n = 28$), near-normal events ($|\theta_m| \leq 5°$) 0.57 ($n = 10$), and events with a positive offset ($\theta_m > +5°$, up to $+25°$ on the opposite side) only 0.45 ($n = 14$). The negative-offset and positive-offset means differ significantly (Welch $t = 7.3$; $p < 10^{-8}$).

This sign-dependent structure is not reproduced by the directional projection of the incident spectrum onto the 1D shore-normal boundary. Integrating a cos-2s directional distribution with the prescribed mean angle and spreading over the onshore half-plane gives the cross-shore energy-flux fraction. It predicts a mean transmission of 0.88, a loss of only about 12%, against the observed mean loss of 43%. That projected transmission is also nearly identical for negative and positive offsets (0.88 versus 0.88) and is itself uncorrelated with the observed K_{bnd} ($r = -0.09$). Directional spreading geometry therefore accounts for at most a small, symmetric fraction of the deficit and cannot explain either its magnitude or its sign-asymmetry. A simple 1D lateral-redistribution argument, in which obliquely incident energy is lost for want of a second horizontal dimension, predicts a dependence on obliquity magnitude that the data do not show, and so is likewise insufficient on its own. The dominant feature is a large, sign-dependent transmission asymmetry superimposed on a substantial direction-independent baseline deficit: even the near-normal events transmit only about 0.55, far below the 0.91 cross-shore directional projection. The flat-bottom diagnostic of Section 3.12 isolates and reproduces both components on a horizontal bed with no shoaling, breaking, or site directional climate. This identifies them as numerical properties of the 1D wave-generation boundary, a loss in the reconstruction of the boundary signal from the directional JONSWAP specification, and not as cross-shore or site-physical effects. The mechanism is discussed further in Section 4.1. Two events, event_0010 and event_0011, exhibit extreme near-boundary wave height collapse under H_{m0}^{BC} values of 4.3 to 4.6 m in approximately 10 m water depth, consistent with individual wave heights in the reconstructed random sea exceeding the depth-limited breaking threshold $H/h \approx 0.6$. This behavior is physically correct for those sea states and is not a numerical error, but it renders the boundary energy transmission for these two events unrepresentative of the broader event set.

Because the boundary energy deficit is direction-dependent and varies up to a factor of two across events, absolute comparisons between modeled and observed H_{m0} or wave energy flux are dominated by the boundary transmission loss and not by cross-shore transformation physics. Absolute model-data comparison metrics are therefore not used in this study. Instead, model-observation comparisons are performed on the three levels defined in Section 2.6: the boundary-referenced ratio K (Eq. (9)), which measures boundary delivery and cross-shore transformation together; the delivered-energy-referenced ratio K_{res}, which isolates the transformation; and the corrected-boundary experiment of Section 3.10. Using the same prescribed H_{m0}^{BC} in the denominator on both sides puts model and observations in the same units, but it does not remove the deficit, because only the modeled side is subject to it. The energy flux ratio Φ is reported qualitatively only (Section 2.3). As stated in Section 2.6, the normalized framework was adopted post-run in response to the deficit discovery; all skill metrics based on it are exploratory.

The phase realization of the boundary signal was not controlled. No random keyword appears in any of the archived parameter files, so XBeach applied its default of random = 1 and drew the seed for the JONSWAP phase reconstruction from the system clock at run time. Prescribing the boundary through a jonstable file fixes the bulk parameters, H_{m0}, T_p, mean direction, peak enhancement, and spreading, but not the phases, which the model reconstructs internally. The runs are therefore not bitwise reproducible, and stochastic variability in the phase realization cannot be excluded as a contributor to the scatter in K_{bnd} or to the near-zero event-scale correlation. Bounding that contribution requires replicate runs at fixed forcing, which the archived simulation set does not contain.

To examine whether the boundary deficit persists for near-normal events retained after the 30° threshold, K_{bnd} was binned by directional deviation in three sub-ranges: 0–10°, 10–20°, and 20–30° from shore-normal. Mean K_{bnd} values were 0.554, 0.607, and 0.570 respectively, and a one-way ANOVA found no statistically significant difference between bins ($p = 0.45$). A one-sample t-test confirmed that the 0–10° bin mean is significantly below 1.0 ($p \approx 0$). The deficit was therefore not primarily driven by residual directional obliquity within the accepted event set but is a systematic property of the 1D JONSWAP boundary even at near-normal incidence (Figs. S1 and S4).

3.3. Event set characteristics

3.3.1. Wave climate and temporal distribution

The 8 m Waverider buoy record for the period 2010 to 2023 shows a highly variable wave climate at the FRF Duck site, with H_{m0} ranging from near-calm conditions below 0.3 m to storm peaks exceeding 4.5 m (Fig. 4a). Energetic conditions with H_{m0} exceeding 2.0 m occur predominantly during the autumn to winter nor'easter season, while quiescent conditions below 1.0 m dominate the summer months and are distributed year-round. The instrument outage from approximately mid-2013 to early 2016 is visible as a data gap in the record. The full study period following restoration of continuous data collection in 2016 shows a higher density of energetic events than the pre-gap period, with several nor'easters reaching H_{m0} above 3.5 m between 2017 and 2023.

Application of the four eligibility criteria to the full record yielded 86,281 candidate windows. Following the stratified random sampling procedure described in Section 2.3.3, a final set of 52 events was selected. The event set is distributed across the full study period (Fig. 4b), with representation in each year of continuous record. Events spanned the full H_{m0} range within each energy class, from near-threshold energetic events at approximately 2.0 m to the largest nor'easter conditions reaching 4.4 m, and from quiescent events near the lower selection threshold of 0.3 m to moderate events approaching 2.0 m. This dynamic range is relevant to the skill assessment in Section 3.4, because it ensures that the relationship between H_{m0} skill and

dissipation skill is assessed across genuinely distinct hydrodynamic regimes and not a narrow cluster of conditions.

3.3.2. Sandbar morphological states

The sandbar state strip in Fig. 4 shows that single-barred conditions (blue) dominated the FRF record throughout the study period, accounting for most of the continuous record both before and after the instrument outage. Unbarred conditions (grey) occurred episodically, concentrated in approximately 2011 to 2013, 2019 to 2020, and portions of 2022 to 2023. Double-barred conditions (purple) were rare, appearing in brief windows around 2015 and again in 2022 to 2023. This distribution reflects the known multi-year sandbar lifecycle at Duck, in which bars migrate offshore over one to three year cycles and occasionally dissipate entirely before reforming, as documented by Ruessink and Kroon (1994), Gallagher et al. (1998), and Plant et al. (1999).

Of the 52 selected events, the stratified sampling design achieved representation across all three sandbar states in each energy class, and every cell target in Table 1 was met exactly: 4 double-barred, 4 unbarred, and 8 single-barred events in each of the energetic and moderate classes, and 4, 8, and 8 respectively in the quiescent class. The achieved counts therefore equal the targets, and no cell was short of eligible candidates. Double-barred conditions are nonetheless sampled at half the density of single-barred conditions because the design set their target at 4 rather than 8, reflecting the rarity of double-barred states in the FRF record. The implications of this deliberate imbalance for statistical conclusions are discussed in Section 4.2.

3.3.3. Cross-shore bathymetric variability

While model simulations and gauge comparisons are restricted to the primary transect at $y_{FRF} = 932$ m (Section 2.4.2), the CRAB survey DEM provides bathymetric data across the full alongshore extent of the survey domain. Nine transects sampled at equal spacing across that extent, $y_{FRF} = -100$, 44, 188, 356, 500, 644, 812, 956, and 1100 m, are used here to characterize alongshore variability. The following description is provided for context and to establish the representativeness of the primary model-data comparison transect; these nine transects are not simulated. The nine transects show a consistent large-scale profile structure across all morphological states, including a subaerial beach crest near 2 m NAVD88 at $x_{FRF} = 50$–100 m, a steep foreshore, and a smooth outer profile beyond approximately $x_{FRF} = 600$ m that deepens to about 8–10 m at the seaward limit of the surveys. The greatest variability occurs over the active bar zone ($x_{FRF} = 100$–500 m). Single-barred profiles exhibit a pronounced bar crest and shoreward trough, double-barred profiles show distinct inner and outer bars separated by an inter-bar trough, and no-bar profiles are characterized by a smoother concave-up shoreface without a distinct crest. As shown by the plan-view morphology in Fig. S6, these transects sample coherent two-dimensional bar configurations rather than isolated one-dimensional sections.

Profile shape also varies systematically alongshore. Lower-y_{FRF} transects show greater bar relief and more complex trough geometry, whereas higher-y_{FRF} transects are smoother and less barred. This pattern is consistent with known spatial variability in the vicinity of the FRF pier and is evident in both the cross-shore sections (Fig. S5) and the representative bathymetric maps (Fig. S6). The model simulations use the $y_{FRF} = 932$ m profile, which is co-located with the permanent gauge array and is most representative of open-coast conditions away from pier influence.

3.4. Cross-shore transformation skill

Fig. 5 shows the normalized cross-shore wave height transformation ratio (Eq. (9)) for the 52-event ensemble, partitioned by energy class. The boundary transmission ratio K_{bnd}, defined in Section 3.2 as K evaluated at the offshore model boundary node, is used throughout this section as the reference for the direction-dependent energy deficit. Individual event profiles are shown as thin lines, the ensemble median as a thick line, and the interquartile range as a shaded band. Observed K values at each gauge position are overlaid as symbols, with marker shape indicating sandbar state.

The mean boundary transmission ratio K_{bnd} is 0.61 for energetic events, 0.61 for moderate events, and 0.51 for quiescent events, with event-by-event ranges of 0.38 to 0.72, 0.39 to 0.76, and 0.40 to 0.71 respectively. As described in Section 3.2, this deficit reflects a structural limitation of the 1D domain. Normalizing by the prescribed boundary height does not remove it from the comparison, because only the modeled side is subject to it (Section 2.6): the modeled profiles begin at K_{bnd} at the seaward end while the observed profiles begin near unity. The $K = 1$ dashed reference line represents the prescribed boundary condition level, not the offshore wave height, and the vertical offset between the modeled and observed profiles at the seaward end is the boundary

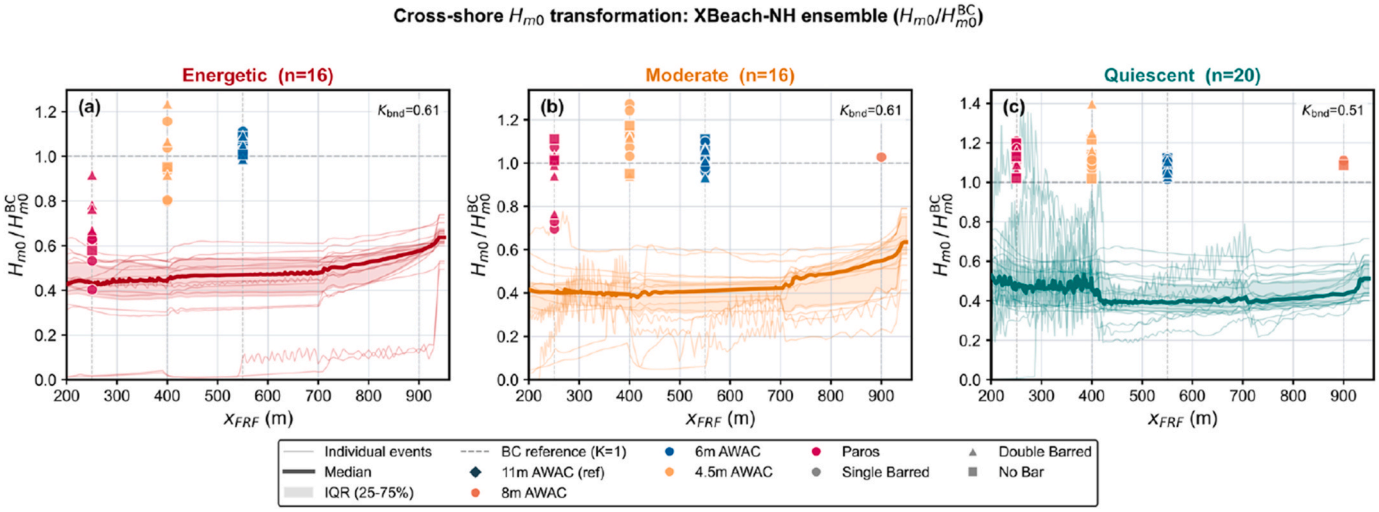


Fig. 5. Cross-shore transformation of significant wave height for the 52-event ensemble, shown as normalized wave height, H_{m0}/H_{m0}^{BC}, where H_{m0}^{BC} is the offshore boundary-condition wave height for each event. Thin lines show individual model realizations, thick lines show the class median, and shading denotes the interquartile range. Symbols show observations at the in-situ gauge locations, grouped by energy class and morphological state. Normalization by the offshore boundary condition removes event-to-event differences in incident wave magnitude and emphasizes the spatial pattern of wave transformation across the surf zone.

deficit itself.

The observed K values revealed a two-stage cross-shore structure consistent with linear shoaling followed by depth-induced breaking. At the 11 m AWAC ($x_{FRF} = 1150$ m), K_{obs} was approximately 1.04 to 1.08, reflecting mild shoaling gain seaward of the bar system. At the 6 m AWAC ($x_{FRF} = 550$ m), K_{obs} reached approximately 1.04 across all energy classes, indicating that the observed waves continue to shoal and gain height relative to the boundary forcing level before breaking is initiated. The 4.5 m AWAC ($x_{FRF} = 400$ m) showed peak K_{obs} values of 1.0 to 1.15, corresponding to the crest of the shoaling amplification immediately seaward of or over the outer bar. Shoreward of this position, breaking dissipation became the dominant process: K_{obs} at the Paros gauge ($x_{FRF} = 250$ m) dropped to 0.66 to 1.13, with the lowest values occurring under energetic conditions (mean $K_{obs} = 0.66$) where breaking over the bar removes a substantial fraction of the incident energy flux before the waves reach the inner surf zone.

The contrast between model and observations is therefore two-layered. First, the model realized only 51 to 61% of the prescribed boundary wave height at the offshore boundary itself, through the wave-generation transmission deficit described in Section 3.2. Second, the observed waves simultaneously gained 4 to 15% through shoaling as they propagated from the boundary depth to the intermediate gauge positions. The net effect is that the modeled K profiles sat systematically below the observed K values at all gauge positions and across all energy classes, with the separation largest at the intermediate gauges where shoaling amplification is strongest.

Despite this systematic offset, the ensemble-median K profiles showed a qualitatively consistent cross-shore shape across energy classes. For energetic events (Fig. 7a), the median modeled K increased gradually from approximately 0.40 at $x_{FRF} = 250$ m to 0.60 at $x_{FRF} = 950$ m, without a clear breaking-induced decrease over the bar zone. For moderate events (Fig. 7b), the median profile was similar in shape with slightly higher values at the inner positions. For quiescent events (Fig. 7c), the ensemble spread was substantially larger than for the other two classes, with individual event K profiles ranging from near zero to above 1.4 in the bar zone. This elevated variance reflects the sensitivity of the normalized ratio to the low H_{m0}^{BC} denominator in quiescent conditions, where small absolute differences in modeled wave height produce large relative differences in K.

The absence of a clear breaking-induced K decrease in the modeled profiles over the bar zone, particularly for energetic events, suggests that XBeach-NH does not reproduce the observed onshore-to-offshore gradient in wave height attenuation correctly at this site and under this boundary condition configuration. This finding is examined further through the individual event comparisons in Fig. S7.

Fig. 6 shows the normalized scatter of modeled versus observed K at all gauge positions for the 52-event ensemble, partitioned by energy class. Each point represents one gauge-event pair, with marker shape indicating sandbar state.

Across all three energy classes the model systematically underpredicts K relative to observations. The bias was −0.637 for quiescent events (n = 40), −0.633 for moderate events (n = 34), and −0.479 for energetic events (n = 36). The smaller bias for energetic events was consistent with the higher K_{bnd} values for that class (mean 0.61 versus 0.51 for quiescent), which reduces the normalized gap between model and observations at the boundary.

The Pearson correlation between modeled and observed K was $r = 0.05$ for quiescent events (n = 40), $r = 0.05$ for moderate events (n = 34), and $r = 0.00$ for energetic events (n = 36). These values are computed on the in-domain pair set defined in Section 2.6 ($n = 110$ pairs over 52 events). Under the event-block bootstrap, the 95% confidence interval on r spans zero in every class (energetic: −0.23 to 0.30, $p = 0.96$; moderate: −0.20 to 0.27, $p = 0.74$; quiescent: −0.15 to 0.29, $p = 0.64$), as it does for the three classes pooled ($r = 0.08$, −0.05 to 0.20, $p = 0.23$). These intervals are wider than the Fisher z intervals that a pairwise-independent treatment would give, because resampling whole events respects the fact that gauges within an event share one boundary realization. The near-zero result therefore rests on the weaker assumption. These results indicate that the correlation between modeled and observed K is statistically indistinguishable from zero across all energy classes: the model cannot predict which events will show stronger or weaker transformation at any individual gauge position. The model reproduces the ensemble-mean transformation level with a consistent negative bias but has no event-scale predictive skill under this configuration.

Fig. 6 reveals no systematic dependence of model skill on sandbar state. Single-barred, double-barred, and unbarred events are distributed without clustering in the scatter for all three energy classes, indicating that the skill deficit is not primarily driven by morphological complexity but by the boundary condition limitation identified in Section 3.2.

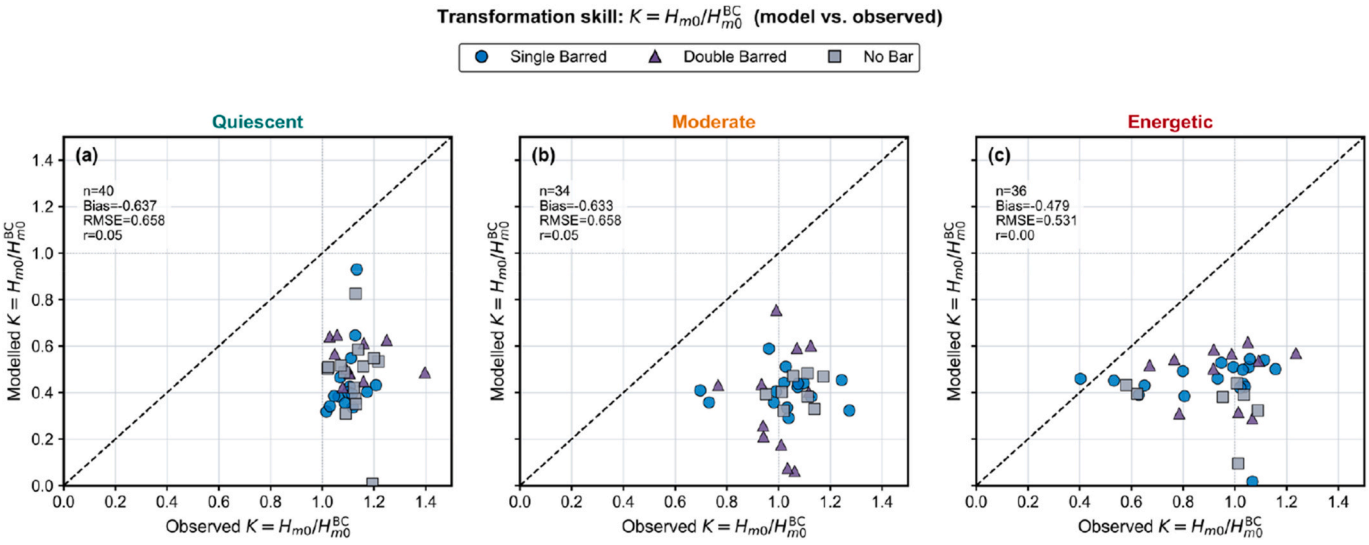


Fig. 6. Skill assessment for normalized wave-height transformation, shown as modeled versus observed $K = H_{m0}/H_{m0}^{BC}$ at the in situ gauge locations for all selected events. Results are grouped by energy class, and marker shape denotes morphological state. The one-to-one line indicates perfect agreement. K is referenced to the prescribed boundary height on both sides, so it measures the combined boundary-delivery and cross-shore-transformation error (Section 2.6); the modeled profiles begin at K_{bnd} rather than at unity. Statistics use the in-domain pair set, $n = 110$.

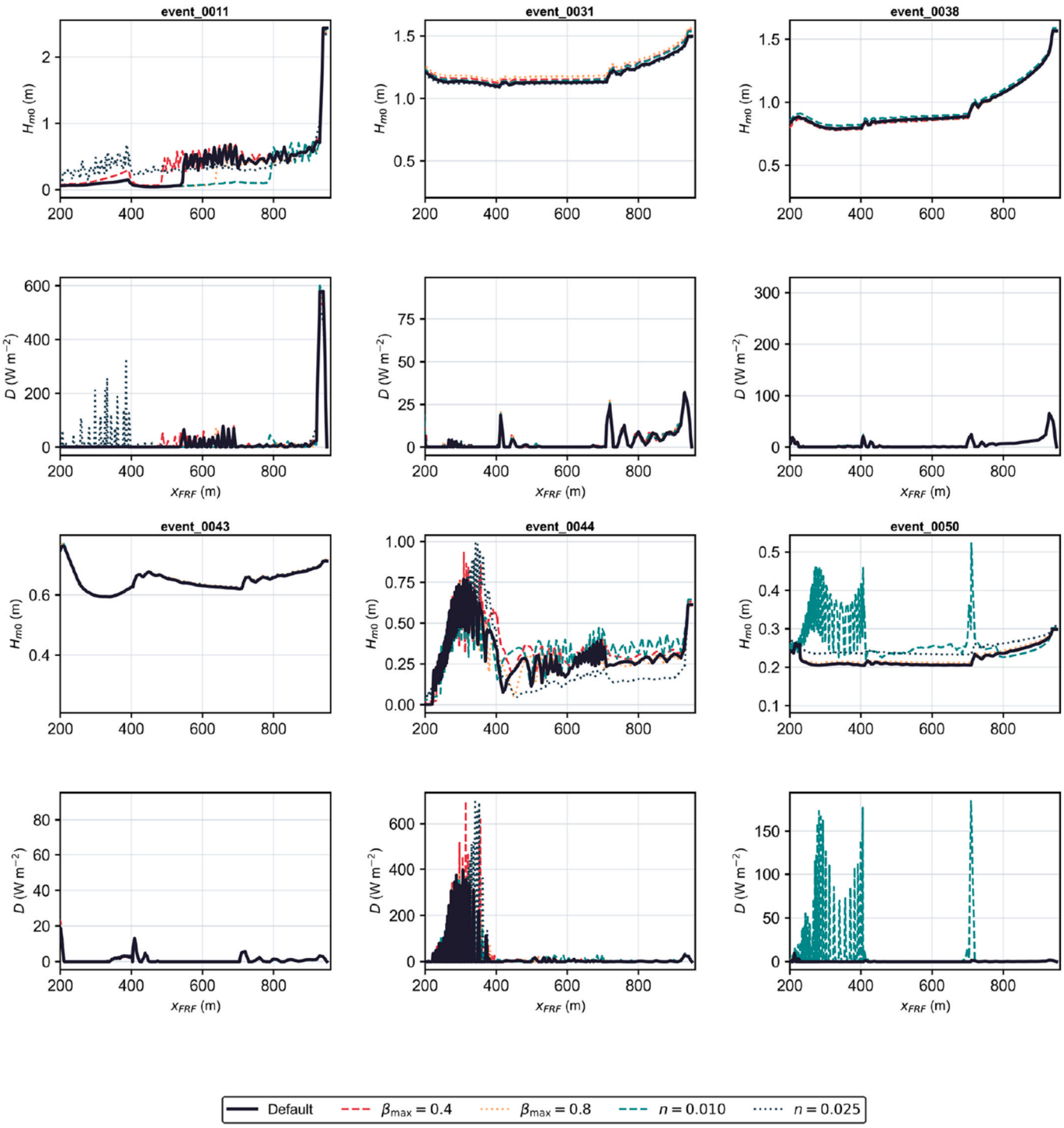


Fig. 7. Sensitivity of modeled cross-shore wave transformation and dissipation to the breaking-steepness parameter β_{max} and Manning's roughness coefficient n, the two parameters active in non-hydrostatic mode, for the six perturbed events: event_0011 and event_0031 (energetic), event_0038 and event_0044 (moderate), and event_0043 and event_0050 (quiescent). For each event, the upper panel shows significant wave height and the lower panel shows dissipation as a function of x_{FRF}, with the default parameterization compared against the perturbed cases.

The observed K values clustered tightly between 0.9 and 1.2 across all energy classes and gauge positions, reflecting the combined effect of shoaling amplification at intermediate gauges and the limited cross-shore attenuation measurable at the five gauge positions. The modeled K values spanned a much wider range from near zero to 0.9, with the lowest values concentrated at the Paros gauge (x_{FRF} = 250 m) under energetic forcing, where the model predicts strong breaking-induced attenuation that is not matched in magnitude by the observations under the normalized framework.

3.5. Transformation skill referenced to delivered energy

The comparison above charges the model for energy its boundary never delivered. Referencing the modeled profile to the energy actually delivered, $K_{res} = K_{model}/K_{bnd}$ (Section 2.6), separates the two error sources and is the appropriate test of the cross-shore transformation

itself. The mean bias contracts from −0.54 to −0.17 and the RMSE from 0.57 to 0.29 on the full valid pair set, but those reductions are arithmetic consequences of dividing by an event-mean constant and are not evidence of skill. The informative quantity is the event-scale correlation, which the rescaling cannot manufacture.

Under that test a weak but resolvable relationship appears where the boundary-referenced comparison showed none. Pooled across the 203 gauge-event pairs the correlation with the observations rises from 0.04 (95% CI −0.06 to 0.13, $p = 0.43$) to 0.21 (0.11 to 0.30, $p < 0.001$) under the event-block bootstrap. The gain is concentrated in the moderate class, which moves from 0.11 (−0.11 to 0.27) to 0.22 (− 0.01 to 0.38), and the energetic class from 0.12 (−0.03 to 0.35) to 0.14 (− 0.01 to 0.34); neither per-class interval excludes zero at $n = 65$ and 63 pairs. The quiescent class remains flat at 0.03 (−0.07 to 0.11), consistent with observed K clustering near unity there with a spread barely above the measurement-noise floor, so there is little transformation signal for any metric to recover.

Referencing the model to the delivered energy changes the interpretation of the null result of Section 3.4. The model carries some event-scale information about cross-shore transformation, which the boundary-referenced comparison masks entirely. The recovered correlation still explains only about 4% of the pair-to-pair variance, so the transformation skill of this 1D configuration is weak rather than absent, and the near-zero skill reported against K overstates the failure by attributing the boundary deficit to the transformation physics.

3.6. Numerical sensitivity and convergence

Grid convergence and spin-up duration were tested for the three representative events introduced in Section 3.1. The parameter sensitivity tests use a separate six-event set and are reported in Section 3.7. All differences reported here are relative to the standard-grid, 30-min spin-up baseline run for each event.

Grid convergence. Coarsening the cross-shore grid from standard to coarse resolution increased the H_{m0} profile RMSE to 0.259 m for event_0010, 0.502 m for event_0038, and 0.121 m for event_0050. The large coarse-grid error for event_0038 reflects the coarser grid's inability to resolve the steeper bathymetric gradients near the double-barred crest. Refining the grid to fine resolution (0.5 m) produced RMSE = 0.310 m for event_0038 and 0.237 m for event_0050 relative to the standard grid, both within the range of the coarse-grid errors. The fine-grid simulation for event_0010 is numerically unstable regardless of CFL reduction: U-velocities exceed the stability threshold at approximately t = 1250 s, consistent with individual wave heights approaching the depth-limited breaking threshold $H/h \approx 0.6$ in 10 m water depth, which generates extreme velocity gradients at sub-metre resolution. Fine-grid convergence for this event therefore cannot be demonstrated with the current model configuration. For the two events for which all three grids are available, the standard grid sits within the envelope bracketed by the coarse and fine results and produces cross-shore K profiles that are structurally identical to the fine-grid solution. These results support the adequacy of the standard grid for the present analysis; the fine-grid instability for the most energetic event is a known limitation of depth-averaged NH models at high forcing levels.

Spin-up duration. Differences between 15-min and 30-min spin-up runs were 0.0220 m mean absolute H_{m0} difference for event_0010, 0.0164 m for event_0038, and 0.00489 m for event_0050. Differences between 60-min and 30-min spin-up runs were 0.0346 m for event_0010, 0.0184 m for event_0038, and 0.00108 m for event_0050. All six differences are at least one order of magnitude smaller than the model-data biases documented in Section 3.4, confirming that 30-min spin-up is sufficient and that the choice of spin-up duration does not materially affect the results.

Breaking steepness and friction sensitivity. The one-at-a-time perturbations of β_{max} and the Manning coefficient are reported in Section 3.7 with Fig. 7, on the six-event set defined in Section 2.5.3. They are not repeated here.

Roller-model parameter sensitivity (γ; α, γ_{max}). As described in Section 2.5.3, these parameters belong to the roller/surfbeat framework and have limited active influence in XBeach-NH nonh mode. For all three events, perturbations to γ, α, and γ_{max} produced mean absolute H_{m0} differences of less than 0.036 m, consistent with their structural insensitivity in this model configuration. These results confirm that the primary breaking control in NH mode is β_{max}, not the roller parameters, and that the central conclusions are robust to the roller-parameter choices. No parameter combination altered the sign or approximate magnitude of the skill deficits identified in Section 3.4.

3.7. Parameter and morphology sensitivity

Fig. 7 shows the sensitivity of modeled cross-shore H_{m0} profiles and depth-integrated dissipation $D(x)$ to the two parameters that are active in non-hydrostatic mode, the breaking steepness threshold β_{max} and the Manning friction coefficient n, for the six perturbed events of Section 2.5.3, two per energy class. The default parameter values are $\beta_{max} = 0.6$ and $n = 0.015$ m $-1/3$ s. Sensitivity runs vary β_{max} over 0.4 to 0.8 and n over 0.010 to 0.025 m $-1/3$ s independently.

For the two quiescent events (event_0043 and event_0050) and the moderate event_0038, the H_{m0} profiles showed minimal sensitivity to both parameters across the full cross-shore extent: the mean absolute difference from the default profile stayed below 0.024 m for every perturbation except the low-n run of event_0050 (0.071 m), which is confined to the inner surf zone. Dissipation profiles were similarly insensitive, with parameter perturbations producing changes within the observational uncertainty range described in Section 2.2.2.

Sensitivity grows with forcing level but remains an order of magnitude below the boundary deficit. The largest responses occur for the most energetic event, event_0011 ($H_{m0}^{BC} = 4.56$ m), where the high-n run departs from the default by 0.165 m in the mean and 0.53 m locally, and for the moderate event_0044, where the corresponding values are 0.116 m and 0.60 m. In both cases the largest local departure falls in the bar zone, at $x_{FRF} = 386$ and 348 m respectively, while at the offshore boundary the same perturbations shift H_{m0} by less than 0.05 m, so the parameters redistribute energy within the profile without altering what the boundary delivers. Which parameter dominates varies between events: friction gives the larger response for event_0011 and event_0044, whereas β_{max} gives the larger response for event_0031, the other energetic event. For comparison, the boundary transmission deficit removes 40% or more of the prescribed offshore height for every event in the set, so no admissible choice of β_{max} or n approaches the magnitude of the model-data discrepancy documented in Section 3.4. No single parameter change reversed the cross-shore structure of the K profile or altered the sign of the boundary-deficit pattern that is the central diagnostic finding of this study.

Fig. S7 illustrates the contrast between model and observations for two individual events: an energetic single-barred event on 2020-09-22 ($H_{m0}^{BC} \approx 3.7$ m) and a quiescent double-barred event on 2022-02-11 ($H_{m0}^{BC} \approx 0.35$ m). These events are selected as representative of the contrasting physical regimes in the dataset and are not the events with best or worst skill.

For the energetic event (Fig. 10a and c), the modeled H_{m0} profile increases gradually from approximately 1.6 m at $x_{FRF} = 250$ m to approximately 2.3 m at $x_{FRF} = 950$ m. This onshore-increasing profile is physically inconsistent with the expected breaking-induced attenuation over the single bar crest visible near $x_{FRF} = 350$ to 400 m in panel (c). The observed gauge values (colored symbols) show H_{m0} values of approximately 1.9 to 3.7 m at the intermediate positions, consistent with shoaling amplification before breaking. The model-observation gap at all gauge positions reflects the K_{bnd} deficit for this event rather than a cross-shore transformation error per se, but the absence of a modeled breaking signature over the bar is a genuine physical deficiency under

this boundary configuration.

For the quiescent event (Fig. 10b and d), the modeled H_{m0} profile is nearly flat at approximately 0.15 m across the full cross-shore extent, with essentially no cross-shore transformation signal. The observed gauge values of approximately 0.35 to 0.37 m are consistently more than double the modeled values at all positions. This event has one of the lowest K_{bnd} values in the quiescent class (approximately 0.40), meaning the model receives approximately 40% of the prescribed boundary energy; the effective offshore H_{m0} at the boundary node is approximately 0.40×0.35 m $= 0.14$ m, fully consistent with the near-flat modeled profile across the cross-shore extent. The double-barred morphology visible in panel (d) produces no discernible signature in the modeled H_{m0} profile, consistent with the absence of active breaking at these low energy levels.

3.8. Hydrostatic baseline comparison

The phase-averaged hydrostatic baseline (Section 2.7) provides a reference for interpreting the near-zero XBeach-NH transformation skill reported in Section 3.4. Both models were scored on the identical set of 203 gauge-event pairs (65 energetic, 63 moderate, 75 quiescent), so the comparison is internally consistent. This 203-pair set comprises every gauge-event pair with a valid observed K; it is larger than the 110-pair subset used for Fig. 6 scatter in Section 3.4, which is restricted to gauges falling inside the model domain. The near-zero XBeach-NH skill is reproduced on both pair sets, so the choice of set does not affect the conclusion. Recomputed on this pair set, XBeach-NH reproduced Section 3.4 result: an ensemble Pearson correlation of $r = 0.04$ (event-block bootstrap 95% CI -0.05 to 0.13) and a mean K bias of -0.54, statistically indistinguishable from zero skill. The hydrostatic baseline, scored identically, returned $r = 0.65$ (95% CI 0.54 to 0.74) and a near-zero bias of -0.02 for the $\gamma = 0.50$ index. The result was insensitive to the breaker index: $\gamma = 0.42$ gave $r = 0.65$ and a bias of -0.05. All confidence intervals in this section and in Table 3, on bias and RMSE as well as on Pearson r, are computed by the event-block bootstrap of Section 2.6 (resampling the 52 events) rather than by the Fisher transformation or a t interval, so they account for the non-independence of the gauge-event pairs sharing one boundary realization within each event. Table 3 and Fig. 8 summarize the per-class skill.

Table 3
Cross-shore transformation skill for $K = H_{m0}/H_{m0}^{BC}$ at the gauge positions, by energy class, for XBeach-NH and the phase-averaged hydrostatic baseline (Battjes-Janssen, $\gamma = 0.42$; Thornton-Guza, $\gamma = 0.50$). n is the number of gauge-event pairs. All 95% confidence intervals, on bias, RMSE, and Pearson r alike, are computed by the single event-block bootstrap of Section 2.6 (resampling the 52 events with replacement, 20{,}000 iterations), so every interval accounts for the non-independence of gauge-event pairs within an event. Both models are scored on the identical pair set.

Energy class	Model	n	Bias [95% CI]	RMSE [95% CI]	r [95% CI]
Energetic	XBeach-NH	65	−0.46 [-0.52, −0.40]	0.50 [0.43, 0.57]	0.12 [-0.03, 0.35]
	BJ ($\gamma = 0.42$)	65	−0.09 [-0.11, −0.07]	0.16 [0.14, 0.18]	0.60 [0.44, 0.71]
	TG ($\gamma = 0.50$)	65	−0.04 [-0.06, −0.01]	0.12 [0.10, 0.15]	0.62 [0.46, 0.75]
Moderate	XBeach-NH	63	−0.54 [-0.59, −0.50]	0.57 [0.52, 0.62]	0.11 [-0.11, 0.27]
	BJ ($\gamma = 0.42$)	63	−0.02 [-0.05, 0.00]	0.09 [0.07, 0.10]	0.49 [0.29, 0.64]
	TG ($\gamma = 0.50$)	63	0.00 [-0.03, 0.02]	0.09 [0.06, 0.11]	0.44 [0.17, 0.66]
Quiescent	XBeach-NH	75	−0.60 [-0.64, −0.56]	0.62 [0.58, 0.65]	−0.02 [-0.14, 0.09]
	BJ ($\gamma = 0.42$)	75	−0.03 [-0.04, −0.01]	0.07 [0.05, 0.09]	0.58 [0.41, 0.73]
	TG ($\gamma = 0.50$)	75	−0.03 [-0.04, −0.01]	0.07 [0.05, 0.09]	0.58 [0.41, 0.73]
All	XBeach-NH	203	−0.54 [-0.57, −0.50]	0.57 [0.53, 0.60]	0.04 [-0.05, 0.13]
	BJ ($\gamma = 0.42$)	203	−0.05 [-0.06, −0.03]	0.11 [0.10, 0.13]	0.65 [0.54, 0.73]
	TG ($\gamma = 0.50$)	203	−0.02 [-0.03, −0.01]	0.10 [0.08, 0.11]	0.65 [0.54, 0.74]

The contrast held across all three energy classes (Table 3). The hydrostatic baseline correlation was $r = 0.62$ (energetic), 0.44 (moderate), and 0.58 (quiescent), with every 95% confidence interval excluding zero, whereas the XBeach-NH correlation was 0.12, 0.11, and -0.02 respectively, with every interval spanning zero. The baseline RMSE was 0.07 to 0.12 across classes, a factor of four to nine smaller than the XBeach-NH RMSE of 0.50 to 0.62, and the baseline bias was within 0.04 of zero in every class against the -0.46 to -0.60 bias of the non-hydrostatic runs.

This comparison is consistent with the near-zero non-hydrostatic skill arising from boundary energy delivery and not from the breaking parameterization, but the comparison is not on an equal footing and must be read with that caveat. The hydrostatic baseline is initialized at the offshore node with the full prescribed boundary energy and is therefore not subject to the transmission deficit by construction (Section 2.7); the XBeach-NH runs lose 39 to 49% of that energy at the boundary through the direction-dependent deficit of Section 3.2. A higher baseline K at any gauge does not by itself demonstrate superior transformation physics, because the baseline receives more useable energy at the inner array than XBeach-NH does. What the comparison does establish is narrower and still useful: a standard depth-limited breaking model, supplied with the correct boundary energy over the same bathymetry, reproduces the observed cross-shore transformation with genuine event-scale skill, so the observed transformation at this site is well described by depth-limited dissipation and does not require resolved non-hydrostatic dynamics to be captured. It does not isolate the non-hydrostatic breaking physics, which a clean test would require by propagating the same deficient boundary energy through both a hydrostatic and a non-hydrostatic solver; the in-model surfbeat ablation (Section 3.9) is a step toward that test but is limited to three events. The inference that a boundary-corrected non-hydrostatic configuration, for example a 2DH domain (Section 4.1), would recover comparable transformation skill is therefore a hypothesis motivated by this contrast, not a demonstrated result.

3.9. In-model hydrostatic ablation

The parametric baseline of Section 3.8 is forced at the offshore node with the full boundary energy and therefore cannot, by construction, reveal whether the boundary transmission deficit of Section 3.2 is a property of the non-hydrostatic solver or of the 1D boundary itself. The in-model hydrostatic ablation (Section 2.8) addresses this directly by propagating the identical boundary forcing through XBeach in hydrostatic surfbeat mode on the same grid and bathymetry. Table 4 reports the boundary transmission ratio K_{bnd} for the two wave models across the three representative events, together with the comparison figure (Fig. 9).

This ablation is a three-event proof of concept, not an ensemble result, and the energetic case (event_0010) is the same near-boundary-collapse event flagged in Section 3.2 as unrepresentative; the values below should be read accordingly. The hydrostatic run exhibits a boundary deficit for every event: K_{bnd} is 0.75, 0.89, and 0.53 for the energetic, moderate, and quiescent cases respectively, all below unity. The non-hydrostatic values, 0.38, 0.76, and 0.49, reproduce the Supplementary Table S1 entries for these events under the same first-node boundary definition used throughout (Section 3.2). The boundary energy loss is therefore not unique to the non-hydrostatic solver. A 1D domain forced with a directional jonstable boundary delivers less than the prescribed energy to the model interior in the hydrostatic model as well, so a component of the deficit is a property of the 1D

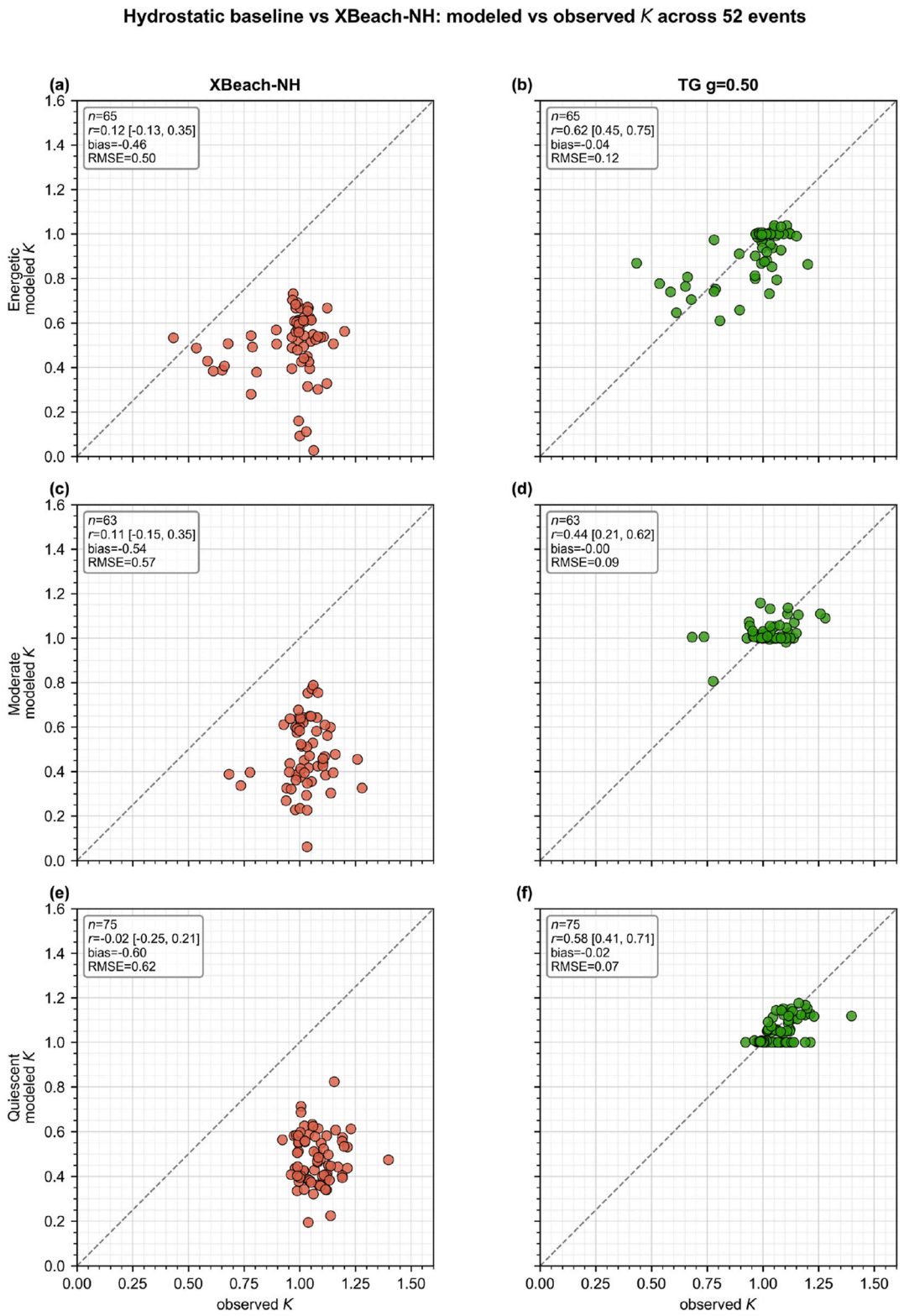


Fig. 8. Modeled versus observed normalized transformation ratio $K = H_{m0}/H_{m0}^{BC}$ at the gauge positions for all 52 events, grouped by energy class (rows) for XBeach-NH (left) and the Thornton-Guza hydrostatic baseline ($\gamma = 0.50$, right). Each point is one gauge-event pair, and the dashed line is the one-to-one relation. Annotations give the number of pairs, the Pearson correlation with its 95% confidence interval, the mean bias, and the RMSE. The non-hydrostatic points fall well below the one-to-one line with no correlation, whereas the hydrostatic baseline clusters on the one-to-one line with significant event-scale skill.

boundary configuration shared by both wave models.

In all three events the non-hydrostatic deficit is larger than the hydrostatic one, and across these three cases the gap increases with sea state. The transmission ratio separates into a component shared by both wave models, $1 - K_{bnd}^{SB}$, and a non-hydrostatic-specific component, $K_{bnd}^{SB} - K_{bnd}^{NH}$ (Table 4). For the quiescent event the deficit is almost entirely shared (0.47 of a total 0.51) and the non-hydrostatic-specific loss is negligible (0.04). For the energetic event the relation inverts:

Table 4
Boundary transmission ratio $K_{bnd} = H_{m0,model\ boundary}/H_{m0}^{BC}$ for the three representative events in XBeach-NH (non-hydrostatic) and XBeach surfbeat (hydrostatic) mode, with the deficit decomposed into a component shared by both wave models ($1 - K_{bnd}^{SB}$) and a non-hydrostatic-specific component ($K_{bnd}^{SB} - K_{bnd}^{NH}$). Both runs use the identical grid, bathymetry, and jonstable boundary forcing.

Event (class)	H_{m0}^{BC} (m)	K_{bnd} XBeach-NH	K_{bnd} surfbeat	Shared deficit $1 - K_{bnd}^{SB}$	NH-specific $K_{bnd}^{SB} - K_{bnd}^{NH}$
Event 0010 (energetic)	4.25	0.38	0.75	0.25	0.37
Event 0038 (moderate)	2.00	0.76	0.89	0.11	0.13
Event 0050 (quiescent)	0.60	0.49	0.53	0.47	0.04

the shared component is modest (0.25) and the non-hydrostatic-specific loss is the larger contribution (0.37), reducing the transmitted height to 0.38 of the prescribed value. Because the energetic case is the flagged collapse event, this inversion should be regarded as illustrative rather than as an established scaling. Fig. 9a makes this visible: the hydrostatic profile (blue) holds near 3 m at the boundary and shoals smoothly shoreward, whereas the non-hydrostatic profile (red) collapses and oscillates within the first 200 m. This energy-dependent splitting is consistent with the depth-limited near-boundary breaking documented for the most energetic events in Section 3.2, in which individual wave heights in the reconstructed sea state exceed the breaking threshold $H/h \approx 0.6$ in approximately 10 m depth, a mechanism that acts on the phase-resolved non-hydrostatic field but not on the phase-averaged surfbeat envelope.

Within the limits of a three-event test, the ablation refines rather than overturns the boundary-deficit interpretation. A component of the deficit, dominant under low-energy forcing in these cases, is a shared limitation of the 1D directional boundary that affects both wave models.

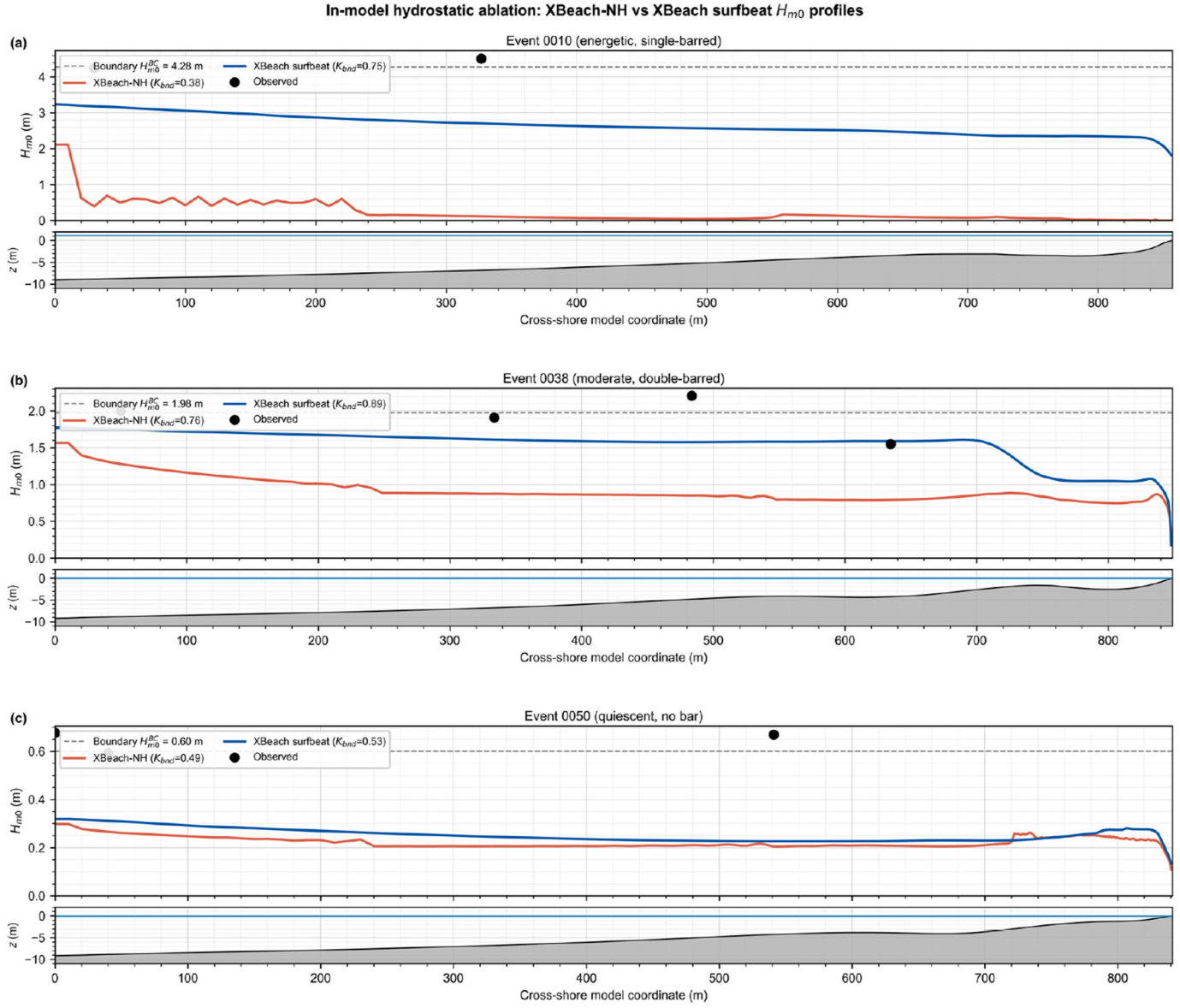


Fig. 9. In-model hydrostatic ablation comparing XBeach-NH (non-hydrostatic, red) and XBeach surfbeat (hydrostatic, blue) cross-shore H_{m0} profiles for the three representative events: (a) energetic event_0010, (b) moderate event_0038, and (c) quiescent event_0050. Each panel shows the two modeled profiles against the prescribed boundary H_{m0} level (dashed) and the observed gauge values (symbols), with the bed profile beneath. The hydrostatic profile lies below the boundary level in every case, confirming a deficit shared by both wave models, but the non-hydrostatic profile falls further still, with the gap largest for the energetic event where near-boundary wave height collapse is most severe.

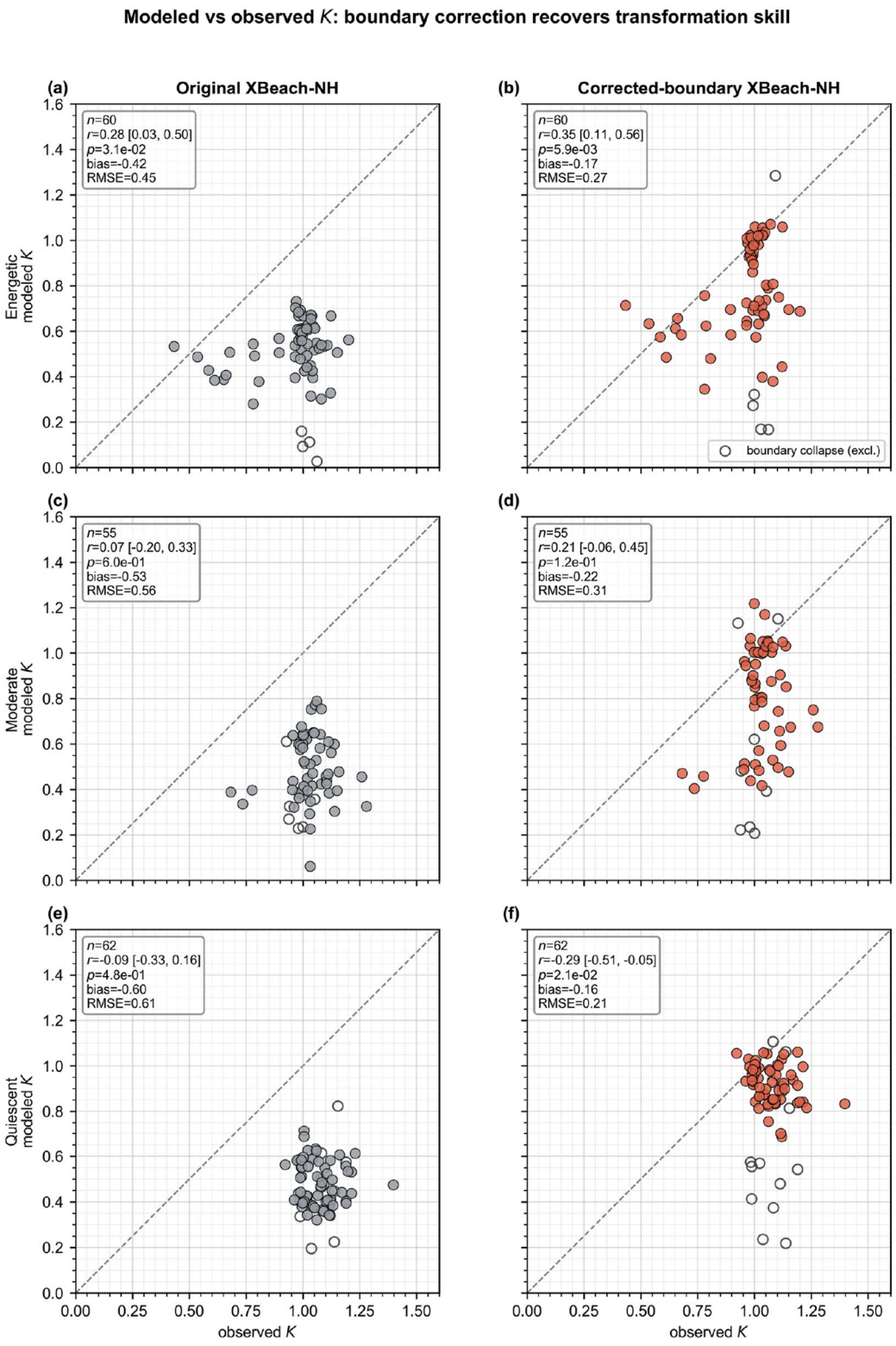


Fig. 10. Modeled versus observed normalized transformation ratio $K = H_{m0}/H_{m0}^{BC}$ at the gauge positions, grouped by energy class (rows) for the original 1D XBeach-NH runs (left) and the corrected-boundary runs (right) in which the prescribed boundary H_{m0} is inflated by $1/K_{bnd}$ per event. Each filled point is one gauge-event pair, and the dashed line is the one-to-one relation. Open symbols mark the energetic events whose inflated forcing collapsed at the offshore boundary; these are excluded from the panel statistics. Annotations give the number of pairs, the Pearson correlation with its 95% confidence interval, the p-value, the mean bias, and the RMSE. The original points fall well below the one-to-one line with no correlation, whereas the corrected points lift toward the line with significant skill in the energetic class and a reduced bias throughout.

The remainder, larger for the energetic case, is a non-hydrostatic-specific near-boundary loss that the depth-limited near-boundary breaking of Section 3.2 would produce. Both components reduce the energy delivered to the surf zone without implicating the cross-shore breaking physics, which Section 3.8 shows is well described once the boundary energy is supplied. A solver-specific loss that grows with sea state would be consistent with the energy-class structure of the skill deficit in Section 3.4, where the normalized model-data gap is largest for the most energetic events, but establishing that scaling would require extending the ablation beyond the present three events.

3.10. Corrected-boundary controlled comparison

The hydrostatic baseline of Section 3.8 and the in-model ablation of Section 3.9 both implicate the boundary energy delivery and not the cross-shore breaking physics, but neither isolates the non-hydrostatic solver under correct forcing: the baseline runs a different solver, and the ablation supplies the deficient boundary to both. The controlled test is to deliver the intended offshore energy to the same non-hydrostatic solver. Because the boundary transmission ratio K_{bnd} is known for each event (Section 3.2, Supplementary Table S1), the prescribed boundary H_{m0} can be inflated by the per-event factor $1/K_{bnd}$ so that the energy reaching the offshore nodes matches the originally intended target. All other settings, period, direction, directional spreading, bathymetry, tide, wind, and every numerical parameter, are held identical to the baseline runs. The entire 52-event set was re-run in this corrected-boundary configuration and scored against the same gauge-event pairs and observed K used in Section 3.8.

The inflation recovers the target offshore energy for the great majority of events: the delivered first-node transmission ratio rises from a sample mean of 0.57 to approximately unity, closing the deficit that the uncorrected runs could not. The exception is a set of seven energetic, low-K_{bnd} events for which the required inflation factor (up to 2.6) drives the prescribed boundary H_{m0} to between 3 and 11 m. At the roughly 9 m offshore boundary these heights exceed the depth-limited breaking threshold, and the reconstructed sea state breaks within the first two to three model cells, so the corrected forcing is not validly delivered. These events are identified by a sharp near-boundary height drop (a ratio of the first-node height to the near-offshore plateau exceeding 1.5) and are excluded from the corrected statistics; the same near-boundary breaking mechanism, amplified by inflation, that Section 3.2 documents for the most energetic uncorrected events. The result below is robust to this screening: including all 52 events the corrected skill is $r = 0.22$ ([0.11, 0.33], $p = 5 \times 10^{-4}$, $n = 203$), and excluding the seven collapse events it is $r = 0.30$ ([0.16, 0.41], $p = 1 \times 10^{-3}$, $n = 177$). These confidence intervals and p-values are computed by an event-block bootstrap that resamples the 52 events, so they account for the non-independence of the gauge-event pairs within an event; the recovered skill remains significant under this cluster-robust test, without relying on a pair-wise independence assumption.

Supplying the correct offshore energy transforms the skill from null to significant (Table 5, Fig. 10). Pooled across all gauge-event pairs, the mean bias is roughly halved, from −0.54 to −0.18, the RMSE falls from 0.57 to 0.26, and the Pearson correlation rises from 0.04 (indistinguishable from zero) to 0.30. The recovery is concentrated in the energetic class, where the correlation rises from 0.12 to 0.35 ($p = 3 \times 10^{-4}$) and the bias contracts from −0.46 to −0.17; the moderate class improves more modestly (0.11 to 0.21, not significant under the event-block bootstrap at $n = 55$). The quiescent class is the exception: its correlation moves from near zero to a statistically significant but physically uninterpretable −0.29 ($p = 0.005$). This anti-correlation reflects an absence of resolvable signal rather than genuine negative skill. Across the quiescent events the observed K values cluster near unity with a standard deviation of only 0.08, marginally above the measurement-noise floor of approximately 0.065 (6% of the mean K). The event-to-event ranking that the correlation measures is therefore dominated by gauge noise, and inflating the boundary by $1/K_{bnd}$ amplifies that noise without a transformation signal to recover.

Table 5
Normalized transformation skill for $K = H_{m0}/H_{m0}^{BC}$ at the gauge positions, for the original 1D XBeach-NH runs and the corrected-boundary runs in which the prescribed boundary H_{m0} is inflated by $1/K_{bnd}$ per event. The corrected rows exclude the seven energetic events whose inflated forcing collapsed at the offshore boundary, except the "Corrected (all 52)" row, which retains them to show robustness. All 95% confidence intervals, on bias, RMSE, and Pearson r alike, come from the single event-block bootstrap of Section 2.6 (resampling the 52 events, 20{,}000 iterations), so every interval accounts for within-event non-independence. n is the number of gauge-event pairs.

Energy class	Configuration	n	Bias [95% CI]	RMSE [95% CI]	r [95% CI]
Energetic	Original	65	−0.46 [-0.52, −0.40]	0.50 [0.43, 0.57]	0.12 [-0.03, 0.35]
	Corrected	60	−0.17 [-0.23, −0.13]	0.27 [0.21, 0.33]	0.35 [0.19, 0.52]
Moderate	Original	63	−0.54 [-0.59, −0.50]	0.57 [0.52, 0.62]	0.11 [-0.11, 0.27]
	Corrected	55	−0.22 [-0.28, −0.16]	0.31 [0.26, 0.36]	0.21 [-0.19, 0.39]
Quiescent	Original	75	−0.60 [-0.64, −0.56]	0.62 [0.58, 0.65]	−0.02 [-0.14, 0.09]
	Corrected	62	−0.16 [-0.18, −0.13]	0.21 [0.18, 0.23]	−0.29 [-0.44, −0.11]
All	Original	203	−0.54 [-0.57, −0.50]	0.57 [0.53, 0.60]	0.04 [-0.05, 0.13]
	Corrected (all 52)	203	−0.22 [-0.26, −0.18]	0.32 [0.28, 0.37]	0.22 [0.11, 0.33]
	Corrected (valid)	177	−0.18 [-0.21, −0.15]	0.26 [0.23, 0.30]	0.30 [0.16, 0.41]

The corrected comparison supports the central interpretation of this study, that boundary energy delivery, and not the breaking parameterization, was the dominant limitation of the uncorrected 1D configuration. It is a sensitivity experiment, because the inflation factor applied to each event is derived from the run being corrected. The out-of-sample test of Section 3.11 addresses that circularity by predicting the inflation from the prescribed wave direction and scoring only withheld events. Once the intended offshore energy is supplied, the same non-hydrostatic solver reproduces the cross-shore transformation with modest event-scale skill. Agreement remains incomplete: a residual negative bias of about −0.18 indicates that the model still systematically under-predicts the transformation ratio, and the recovered correlation accounts for under one-tenth of the pair-to-pair variance. The boundary correction therefore removes most of the apparent skill gap but does not render the 1D non-hydrostatic transformation quantitatively interchangeable with the observations, and for the most energetic events it cannot be applied at all without the offshore boundary itself breaking.

3.11. Out-of-sample test of the boundary correction

The correction applied above uses the K_{bnd} realized by each run, so it is an a posteriori diagnostic rather than a prediction: it requires the run whose deficit it corrects. A correction with predictive value must be built from information available before the run and must be scored on events it has not seen. Section 3.2 supplies the necessary relation, since K_{bnd} depends on the signed directional offset of the boundary forcing, which is prescribed rather than computed. Fitting that relation by ordinary least squares over the 52 events gives $K_{bnd} = 0.550 - 0.0065\,\theta_s$, with θ_s

the signed offset from shore normal in degrees, explaining 73% of the between-event variance in K_{bnd}.

This relation was cross-validated by leaving each event out in turn, refitting on the remaining 51 events, and predicting the withheld event's K_{bnd} from its forcing direction alone. Out-of-sample prediction is nearly as good as the in-sample fit ($R^2 = 0.72$, RMSE 0.058, median absolute error 0.024), and repeated four-fold cross-validation stratified by energy class gives the same result. Adding H_{m0}^{BC} as a second predictor does not improve out-of-sample performance ($R^2 = 0.73$), so the parsimonious direction-only relation is retained.

Applying the withheld-event predictions as a correction, $K_{model}/\widehat{K}_{bnd}$, recovers essentially the same skill as the realized-K_{bnd} correction. On the full valid pair set the pooled correlation with the observations rises from 0.04 (95% CI −0.06 to 0.13) uncorrected to 0.20 (0.11 to 0.29, $p < 0.001$) out of sample, against 0.21 (0.11 to 0.30) for the realized-K_{bnd} correction that uses the answer. The mean bias contracts from − 0.54 to −0.16 out of sample, against −0.17 in sample, and the RMSE from 0.57 to 0.29, against 0.29. Because the out-of-sample and in-sample corrections agree to within their confidence intervals on all three statistics, the recovery reported in Section 3.10 does not depend on knowledge of the run being corrected. The boundary deficit is predictable from the prescribed forcing direction, so the correction can be applied without the run it corrects.

Prediction errors were largest for the most energetic events. The three largest out-of-sample errors are the two most energetic events in the set, event_0010 ($H_{m0}^{BC} = 4.25$ m, $K_{bnd} = 0.38$ observed against 0.62 predicted) and event_0011 (4.56 m, 0.41 against 0.59), together with one moderate event, event_0044 (0.39 against 0.61). In each case the direction-only relation substantially over-predicts the delivered energy, so the correction under-inflates the boundary. These are the same events for which the realized correction drives the boundary past depth-limited breaking (Section 3.10). Direction predicts the boundary deficit well across the bulk of the event set and poorly for the largest sea states, which is a further reason to treat the energetic tail as outside the range where the 1D configuration can be corrected at all.

3.12. Origin of the boundary deficit: flat-bottom diagnostic

A control that prescribes the boundary signal directly, bypassing the directional spectrum, localises the deficit further. A monochromatic wave was imposed as a boundary time series (*wbctype* = *ts_nonh*), with sinusoidal surface elevation and the depth-averaged linear progressive-wave velocity, on the same flat bottom and with the same solver settings. It is delivered essentially in full: the wave height at the generation node is 1.033, 1.001 and 1.004 times the prescribed height at $T = 6$, 10 and 14 s. Under jonstable forcing the same configuration delivers 0.42 to 0.59. The absorbing-generating boundary, the non-hydrostatic solver and the propagation are therefore all capable of carrying the prescribed energy, and the loss occurs when the boundary signal is reconstructed from a directional JONSWAP specification. Narrowing the frequency spectrum alone does not change the delivered height (0.445 at $\gamma_{jsp} = 3.3$ against 0.452 at $\gamma_{jsp} = 20$, both at $s = 10$), which points to the directional rather than the frequency part of that reconstruction. Refining the directional grid from the default 10° resolution to 1° leaves the dependence on wave angle unchanged, so the mechanism is not simply the coarseness of the directional bins.

Because the production runs store only 30 s-averaged free-surface variance, the frequency structure of the deficit could not be examined from the archived output. The flat-bottom configuration was therefore rerun writing instantaneous surface elevation every 0.5 s over a 1-h analysis window, and the delivered variance density spectrum at the offshore nodes was compared with the analytic JONSWAP spectrum prescribed at the boundary. The shortfall is broadband. Across 0.04 to 0.20 Hz, which contains nearly all the variance at a 10 s peak period, the delivered fraction of prescribed variance density lies between 0.29 and 0.36 at shore-normal incidence and between 0.16 and 0.19 at $\theta_m = +20°$, with no systematic trend across the band. The highest band, 0.20 to 0.33 Hz, is delivered better than the rest at the generation node (0.57 and 0.36) and is then damped within the domain, falling to about 0.10 and 0.07 by 100 m inshore, which is the expected fate of short waves in 12 m of water and is distinct from the generation problem. A dispersion or resolution limitation would remove energy preferentially at high frequency, which is not what the spectra show.

The flat-bottom experiment of Section 2.10 isolates the wave-generation boundary from the bathymetry and breaking physics, and it reproduces the boundary deficit in full (Fig. 11). The flat-bottom K_{bnd} and the production K_{bnd} are computed with the identical three-node definition of Section 3.2, in both cases the mean over the wave-generation node and the two adjacent interior nodes divided by the prescribed H_{m0}^{BC}, so the two data sets overlaid in Fig. 11 are directly comparable. The only difference is the node spacing, 5 m on the uniform flat-bottom grid against 10 m in the offshore zone of the production grid, so the averaging window is 10 m rather than 20 m. On a horizontal shelf with no shoaling gradient this has no material effect: at every swept angle the flat-bottom K_{bnd} changes by at most 0.006 between a one-node and a three-node average, and by at most 0.013 between a three-node and a ten-node average, against a deficit of 0.4 or more. At shore-normal incidence ($\theta_m = 0°$), with no shoaling, no breaking, and no directional obliquity, the 1D jonstable wavemaker delivers only $K_{bnd} = 0.58$, close to the production near-normal value of 0.55 (Section 3.2) and far below the analytic cross-shore directional projection of 0.91. This deficit is present at the boundary node itself, is uniform across the flat shelf, and is insensitive to directional spreading: narrowing the spectrum from $s = 10$ to a near-unidirectional $s = 1000$ leaves K_{bnd} near 0.54. It is equally insensitive to the wave-generation settings that might otherwise be suspected of under-resolving the forcing: across boundary time steps of 0.25 to 1.0 s, first- and second-order generation, the boundary-condition version, spectral versus non-spectral generation, and near-monochromatic forcing, the shore-normal K_{bnd} remains within 0.55 to 0.60 (mean 0.57; Supplementary Table S2). The baseline deficit is therefore not a removable configuration choice but an intrinsic

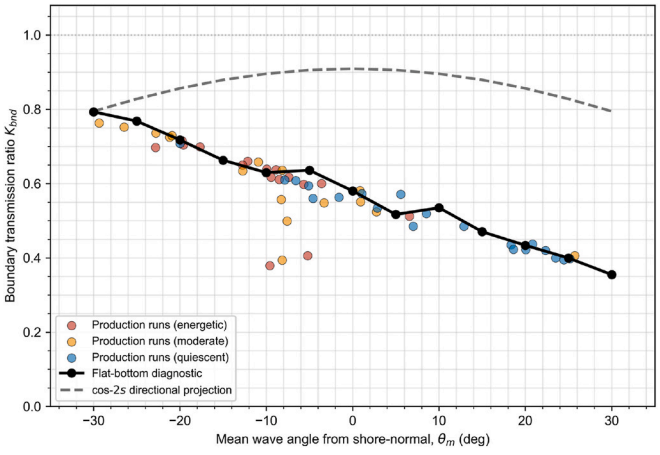


Fig. 11. Flat-bottom boundary-transmission diagnostic. Boundary transmission ratio K_{bnd} as a function of mean wave angle from shore-normal θ_m. The black line is the flat-bottom 1D XBeach-NH diagnostic (Section 2.10), in which bathymetry, shoaling, and depth-limited breaking are removed so that only the wave-generation boundary acts; the dashed grey line is the analytic cos-2*s* cross-shore directional projection of the incident spectrum; colored symbols are the production 52-event K_{bnd} values by energy class. The flat-bottom diagnostic reproduces both the direction-independent baseline deficit (near 0.55 at $\theta_m = 0°$, well below the projection's 0.91) and the monotonic sign-dependent decline of the production data, demonstrating that the boundary deficit is a property of the 1D wave-generation boundary rather than of the cross-shore bathymetry, breaking physics, or site directional climate.

property of how this single-layer non-hydrostatic jonstable boundary delivers energy to the model interior, and the directional projection of the incident spectrum is not its source. It is an intrinsic under-delivery of the jonstable directional wave-generation boundary, and it is the origin of the systematic negative K bias reported throughout (Tables 3 and 5).

Sweeping the mean angle reproduces the sign-dependent structure of Section 3.2 with near-perfect fidelity. On the flat bottom K_{bnd} decreases monotonically from 0.79 at $\theta_m = -30^\circ$ to 0.35 at $\theta_m = +30^\circ$ (Pearson $r = -0.994$), tracking the production 52-event K_{bnd} values across the entire angular range (Fig. 11) and matching the production sign asymmetry (flat-bottom negative-offset mean 0.71, positive-offset mean 0.44; production 0.63 and 0.45). Because the flat-bottom domain contains no sandbars, no refraction, and no seasonal or storm-track covariation, this confirms that the direction dependence of K_{bnd} is a property of how the absorbing-generating boundary handles obliquely specified forcing in a single horizontal dimension, not a cross-shore or site-physical effect. The diagnostic therefore resolves the mechanism left open in Section 3.2: the boundary deficit is the sum of a large direction-independent under-delivery by the jonstable wavemaker and an orientation-dependent modulation of the absorbing-generating boundary under oblique *mainang*, both of which are numerical properties of the 1D wave-generation configuration and not of the resolved cross-shore dynamics. This is consistent with the corrected-boundary result of Section 3.10: once the prescribed energy is restored at the boundary by the $1/K_{bnd}$ inflation, the same solver recovers event-scale transformation skill, precisely because the quantity being corrected is a generation artifact and not a failure of the cross-shore breaking physics.

Moving the offshore boundary into deeper water does not resolve the deficit. Across shelf depths of 9, 12, 17, and 25 m, a near-threefold change, the shore-normal K_{bnd} rises only from 0.553 to 0.595 (0.553, 0.579, 0.585, 0.595), and the $+20^\circ$ value spans 0.416 to 0.465 without a monotonic trend (0.426, 0.416, 0.440, 0.465). The dependence on depth is weak and an order of magnitude smaller than the deficit itself, which remains 0.40 or more at every depth tested, and the spread across three phase realizations at each depth (standard deviation 0.005 to 0.029) covers a substantial part of it. The deficit is therefore not a consequence of boundary placement relative to the breaking zone, and it cannot be removed by extending the domain seaward. This also excludes the possibility that the near-boundary wave height collapse documented for the most energetic events in Section 3.2 is the general mechanism, since that collapse is depth-limited whereas the baseline deficit is not.

The single-layer dispersion limit is likewise not the mechanism, and the test that shows this was designed so that it could have shown the opposite. If the deficit were a dispersion limitation of the depth-averaged single-layer formulation, it would have to grow with kh and to shrink when the reduced two-layer formulation of de Ridder et al. (2021) is enabled. Neither holds. The two-layer formulation delivers slightly less energy than the single layer at every period tested: 0.317 against 0.358 at $kh = 3.03$, 0.345 against 0.397 at $kh = 2.00$, 0.378 against 0.425 at $kh = 1.49$, 0.392 against 0.410 at $kh = 0.99$, 0.418 against 0.428 at $kh = 0.76$, and 0.429 against 0.460 at $kh = 0.52$, with the same ordering for the energetic case ($H_{m0}^{BC} = 4.0$ m, $T_p = 12$ s, $kh = 0.61$): 0.422 against 0.440. At the three shortest periods the reduction, 0.041 to 0.052, exceeds the phase-realization standard deviation of 0.006 to 0.023 by a factor of two or more. Enabling the formulation with the better dispersion relation therefore makes the deficit marginally worse, which is the opposite of what the dispersion hypothesis requires. There is a modest kh dependence in single-layer mode, from 0.46 at $kh = 0.52$ to 0.36 at $kh = 3.03$, so short-wave resolution contributes something at the deep-water end; it cannot be the explanation, because the deficit is still 0.43 to 0.46 at $kh = 0.5$ to 0.8, well inside the range where the single-layer formulation is accurate.

4. Discussion

4.1. Nature and origin of the boundary condition deficit

The boundary transmission deficit identified in Section 3.2 is the dominant factor controlling model-observation differences throughout this study. Its structure is set out quantitatively there and is not repeated here. This section considers which mechanism produces it. Three candidates can be weighed against the evidence of Section 3.2, and the first two are inconsistent with it. Directional spreading geometry, the projection of the incident spectrum onto a shore-normal 1D boundary, predicts a loss of about 12% against an observed 43%, and predicts a loss that is symmetric in the sign of the offset. A 1D lateral-redistribution limitation, in which obliquely incident energy is lost for want of a second horizontal dimension, predicts a dependence on the magnitude of obliquity that the data do not show. The evidence supports the third candidate, a large direction-independent under-delivery combined with a sign-dependent modulation. No symmetric process can produce a sign-dependent response, which points to the orientation-dependent behaviour of the weakly reflective absorbing-generating boundary (van Dongeren and Svendsen, 1997) under obliquely specified forcing.

The flat-bottom experiment of Sections 2.10 and 3.12 resolves the question directly. On a horizontal bed with no shoaling, no breaking, and no site directional climate, the 1D wave-generation boundary reproduces both components: the direction-independent baseline deficit at shore-normal incidence and the full sign-dependent decline with θ_m, tracking the production values across the entire angular range. This excludes the alternative explanations that require site physics. Depth-dependent refraction, seasonal or storm-track covariation, and JONSWAP spectral-shape mismatch cannot be responsible. The flat-bottom domain contains none of them and still reproduces the deficit in full. The deficit is also insensitive to the boundary time step, the wave-generation order, the boundary-condition version, spectral versus non-spectral generation, and the depth at which the offshore boundary is placed (Section 3.12), so it is not a removable configuration choice. A further control excludes numerical resolution of the directional distribution. Refining the directional grid from the default 10° resolution to 1° leaves the shape of the K_{bnd} curve unchanged. A residual mismatch between the prescribed shore-normal reference and the true coastline orientation would shift the sweep horizontally, but it cannot produce a deficit that is already present at exactly shore-normal incidence.

Two further observations constrain the attribution. The in-model hydrostatic ablation (Section 3.9) shows that a surfbeat run forced with the identical 1D boundary also transmits less than the prescribed energy, so a component of the deficit is shared by both wave models and originates in the 1D boundary, not in the non-hydrostatic solver. The same three-event test suggests an additional solver-specific loss at high energy, but does not establish it. The loss is broadband. Runs writing instantaneous surface elevation at the wave time scale (Section 3.12) show that across the energy-containing band, 0.04 to 0.20 Hz, the delivered fraction of prescribed variance density is nearly uniform, between 0.29 and 0.36 at shore-normal incidence and between 0.16 and 0.19 at $\theta_m = +20^\circ$. A dispersion or resolution limitation would remove energy preferentially from the short waves, and the opposite is seen: the 0.20 to 0.33 Hz band is delivered better than the rest at the generation node and is only damped later, within the domain. Together with the layer-count comparison of Section 3.12, in which the two-layer formulation of de Ridder et al. (2021) delivers slightly less energy than the single layer at every period tested instead of more, this locates the mechanism in the wave generation, not in the propagation or the dispersion relation.

This finding has direct practical implications for any study using XBeach-NH in 1D mode with JONSWAP table boundary conditions at sites with oblique wave climates. At Duck, the mean wave direction is approximately 70 to 80° from true north, corresponding to a shore-normal deviation of 0 to 10° for the most favorably aligned events and

up to 25 to 30° for the most oblique events in the selected set. The K_{bnd} range of 0.38 to 0.76 across these events means that absolute model outputs should be interpreted with caution at any site with non-normal wave incidence unless the boundary transmission ratio has been characterized for the specific event direction and domain geometry.

Two mitigation strategies are suggested for future work, and they address different parts of the problem. The first is to run XBeach-NH in 2DH mode, which admits a second horizontal dimension and a genuinely directional boundary and so can deliver obliquely incident energy that a 1D shore-normal boundary cannot. Nonhydrostatic 2DH field simulations using SWASH, which shares the same depth-averaged nonhydrostatic formulation, have demonstrated good infragravity and cross-shore transformation fidelity at barred sites (Rijnsdorp et al., 2015), indicating the approach is physically viable; it would, however, require alongshore bathymetric extrapolation beyond the CRAB survey domain and substantially greater computational cost. The second concerns the boundary spectrum itself, and its scope is more limited: supplying a measured time series with the correct frequency shape in 1D would improve the spectral content but would not address the directional limitation, which is inherently two-dimensional. Only a directional boundary, in 2DH or quasi-3D, can be expected to raise K_{bnd} toward unity; a 1D spectral boundary alone cannot. Whether a 2DH configuration does so at Duck remains to be demonstrated.

Two-dimensional XBeach-NH simulations were not conducted in this study for two reasons. First, constructing 52 event-specific 2DH grids from CRAB survey DEMs, which cover a limited alongshore extent of approximately 700 m at the FRF, would require substantial bathymetric extrapolation beyond the survey domain to define lateral boundary conditions, introducing uncertainty that would complicate interpretation. Second, the computational cost of 52 2DH runs at 1 m cross-shore resolution was considered disproportionate given that the boundary deficit was identified post-run and the study objective was to characterize that deficit rather than to circumvent it. The 1D results are therefore retained as a diagnostic of boundary condition performance.

4.2. Interpretation of near-zero transformation skill

A potential alternative interpretation of the near-zero correlation values is that they reflect a normalization artifact rather than a fundamental model limitation. If the boundary transmission deficit dominates both modeled and observed K, the modeled K range would be compressed toward K_{bnd}, reducing variance and suppressing correlation with observed K regardless of the cross-shore physics. This was examined by comparing the variance of modeled K ($\sigma^2 = 0.04$ to 0.08 across energy classes) against the variance of observed K ($\sigma^2 = 0.01$ to 0.03). The modeled K variance exceeded the observed K variance by a factor of 2 to 3 across all energy classes, indicating that the model produces more event-to-event spread than the observations. This rules out the specific concern that normalization compresses the modeled range toward K_{bnd} and thereby suppresses the correlation; it does not, by itself, exclude all metric artifacts, because a model whose variance is inflated by boundary-direction effects uncorrelated with the observed transformation would also produce near-zero r. That is in fact the interpretation the evidence supports: the modeled K responds primarily to the direction-dependent boundary deficit while the observed K reflects the physical shoaling and breaking structure at the site, so the two vary largely independently across events. The corrected-boundary rerun (Section 3.10) tests this directly: removing the boundary deficit by inflating the forcing recovers a significant correlation between modeled and observed K (0.04 to 0.30), confirming that the deficit was suppressing a correlation that the resolved physics can otherwise produce.

The near-zero Pearson r values between modeled and observed K across all energy classes ($r = 0.00$ to 0.05) indicate that the model cannot predict which events will show stronger or weaker cross-shore transformation at individual gauge positions. The ensemble-median modeled K profiles are physically coherent and reproduce the qualitative shoaling and breaking structure, so the result is not that the model produces random output. A likely contributing factor is the loss of event-specific information at the boundary: the 1D JONSWAP implementation with random phase realizations draws each event from the same statistical distribution, discarding the measured directional spectrum shape and the wave group structure and associated bound infragravity energy (Battjes et al., 2004) that differentiate individual events. The hydrostatic baseline comparison (Section 3.8) is relevant here, with one caveat: the baseline is supplied with the full prescribed boundary energy whereas XBeach-NH is not, so the two are not on an equal footing and the comparison cannot by itself exonerate the non-hydrostatic breaking physics. What it does show is that a standard depth-limited breaking model, given the correct boundary energy, reproduces the observed transformation with event-scale skill on the identical gauge-event pairs. The controlled test that the baseline could not provide, supplying the correct offshore energy to the same non-hydrostatic solver, is reported in Section 3.10: inflating the boundary forcing by $1/K_{bnd}$ per event recovers significant event-scale skill from the non-hydrostatic model, with the pooled correlation rising from 0.04 to 0.30 and the bias and RMSE roughly halving. This confirms that boundary energy delivery, not the breaking parameterization, was the dominant controlling factor in the uncorrected skill gap. The confirmation is partial: a residual negative bias of about −0.18 remains, the recovered correlation explains under one-tenth of the pair-to-pair variance, and for the most energetic events the correction cannot be applied without the inflated boundary itself breaking. The near-zero skill of the uncorrected 1D configuration is therefore largely, though not entirely, an artifact of the boundary energy deficit rather than a failure of the resolved breaking physics.

This interpretation is consistent with the finding in Section 3.4 that sandbar state shows no systematic influence on K scatter. If the model were correctly resolving the cross-shore transformation physics, single-barred events might show systematically different K patterns from unbarred events, as the bar concentrates breaking into a narrower cross-shore zone. The absence of this signal is consistent with the boundary condition limiting the model before the cross-shore physics can differentiate between morphological states, although with five to six gauge positions and the present sample the study is not powered to detect a morphological signal of the expected magnitude, so this null result is suggestive rather than conclusive.

4.3. Implications for nearshore model assessment practice

This study demonstrates that standard 1D non-hydrostatic model assessment against absolute H_{m0} at field sites with oblique wave climates is susceptible to a systematic and direction-dependent boundary bias (Zijlema et al., 2011; Smit et al., 2013) that can be mistaken for cross-shore transformation skill or error. The normalized framework developed here, based on Eqs. (9) and (10), provides a practical approach to separating boundary transmission effects from cross-shore physics in any 1D model assessment context. The framework requires no modification to the model or observation processing and can be applied retroactively to existing model runs.

The FRF gauge array and the event selection methodology developed here provide a reusable comparison platform for any nearshore wave model. The 86,281 eligible candidate windows identified in this study represent a substantial resource that could support systematic comparisons of different boundary condition implementations, model formulations, or parameter sensitivity studies without requiring new field campaigns.

The delivered-energy-referenced ratio $K_{res} = K_{model}/K_{bnd}$ (Section 2.6, results in Section 3.5) is the metric that separates the boundary deficit from cross-shore transformation error, and the separation it achieves is the central quantitative result of this study. Because K_{bnd} derives from the same model run, the reductions in mean bias (from −0.54 to −0.17) and RMSE (from 0.57 to 0.29) follow arithmetically

from the rescaling and are not evidence of cross-shore skill; only the event-scale correlation is informative, because dividing each event's profile by a constant cannot create a relationship between events that was not already there. Under the event-block bootstrap the pooled correlation rises from 0.04 (95% CI −0.06 to 0.13) to 0.21 (0.11 to 0.30, $p < 0.001$), and the split-sample test of Section 3.11 reproduces that recovery from the forcing direction alone on withheld events (0.20, 0.11 to 0.29). The interpretation is that the boundary deficit, not the cross-shore physics, dominates the apparent skill gap, and that the transformation skill of the 1D configuration is weak rather than absent: the recovered correlation still explains only about 4% of the pair-to-pair variance (Fig. S3).

Several limitations should be acknowledged, in roughly decreasing order of consequence. First, the corrected-boundary rerun (Section 3.10) supplies the correct offshore energy to the same non-hydrostatic solver and identifies boundary energy delivery as the dominant controlling factor. The attribution is nonetheless supported but not closed. A residual negative bias and a modest correlation remain after correction, the deficit's own mechanism is characterized but not resolved (Section 4.1), and the correction cannot be applied to the most energetic events without the inflated boundary breaking, so those events are not tested under corrected forcing. Second, the assessment framework, including the normalized metrics, the energy-class stratification, and the exclusion thresholds, was adopted after the boundary deficit was discovered, so all skill statistics are exploratory rather than confirmatory, and the in-model ablation rests on only three events. Third, the model is run in 1D fixed-bed mode along a single transect at $y_{FRF} = 932$ m, co-located with the gauge array, which suppresses the lateral processes that operate at the site. CRAB surveys also precede each event by a median of about six days and by up to 21, so inter-survey bar migration introduces bathymetric uncertainty into the cross-shore comparison. The 1D choice and the boundary deficit are linked, since wave direction is precisely the dimension a 1D domain cannot represent. Fourth, the spectral character of the deficit has been assessed only in the flat-bottom configuration, where dedicated high-frequency output shows the shortfall to be broadband (Section 3.12); it remains untested across the production event set, whose archived output is 30 s-averaged and cannot resolve it. The five to six gauge positions limit the cross-shore resolution of the observed K and dissipation profiles; and events with unusual directional spectra or rapid storm evolution may be under-represented among the 52 selected from 86,281 eligible candidates. The corrected-boundary rerun reported in Section 3.10 addresses the first of these tests directly; the remaining priority future work is to extend the in-model ablation across the full event set and to repeat the assessment in a 2DH configuration, which would additionally test the most energetic events that the 1D boundary correction cannot reach without breaking.

5. Conclusions

This study presents a 52-event field-scale model-data comparison for XBeach non-hydrostatic (XBeach-NH) at the USACE FRF Duck site using 14 years of open-access gauge array observations. The following conclusions are drawn.

Boundary condition transmission deficit. XBeach-NH in 1D non-hydrostatic mode with a JONSWAP table boundary exhibits a systematic wave-height transmission deficit at the offshore boundary. The mean boundary transmission ratio K_{bnd} is 0.57 across the 52 events, with a pooled cross-class range of 0.38 to 0.76. The deficit is strongly direction-dependent (partial $r = -0.937$ with compass bearing, controlling for H_{m0}; $n = 52$) and, diagnostically, depends on the sign of the directional offset rather than its magnitude. It is not explained by directional spreading geometry, which projects to only about 12% loss, nor by a magnitude-dependent lateral-redistribution argument, since K_{bnd} is uncorrelated with the absolute offset; the deficit comprises a large direction-independent under-delivery of the 1D jonstable wavemaker plus a sign-dependent modulation of the absorbing-generating boundary under oblique forcing. A flat-bottom diagnostic that removes all bathymetry, shoaling, and breaking reproduces both components in full ($K_{bnd} = 0.58$ at shore-normal incidence and $r = -0.994$ with wave angle), identifying the deficit as a numerical property of the 1D wave-generation boundary, not of the cross-shore physics or the site wave climate (Sections 3.12 and 4.1). In practical terms, until this deficit is resolved, absolute H_{m0} from 1D XBeach-NH at sites with non-normal wave incidence should not be interpreted without first characterizing the boundary transmission for the specific wave direction and domain geometry.

Post-hoc normalized framework. The boundary deficit was identified post-run, and the normalized transformation ratio $K = H_{m0}/H_{m0}^{BC}$ was adopted in response. All skill assessments using K are therefore exploratory rather than confirmatory. Normalizing by the prescribed boundary height does not remove the deficit from the comparison, because only the modeled side is subject to it: the observed profile begins near unity while the modeled profile begins at K_{bnd}. K is accordingly a measure of boundary delivery and cross-shore transformation combined. The systematic negative bias it shows, with per-class means of −0.48 to −0.63, is of the same magnitude as $1 - K_{bnd}$ and is therefore mostly boundary under-delivery. Referencing the model to the energy its boundary actually delivered, $K_{res} = K_{model}/K_{bnd}$, is the appropriate test of the transformation itself; under it the pooled event-scale correlation rises from 0.04 (95% CI −0.06 to 0.13) to 0.21 (0.11 to 0.30). The cross-shore transformation skill of this configuration is therefore weak rather than absent, and rescaling cannot undo the nonlinear consequences of having propagated deficient energy through breaking, setup, and infragravity generation. Ensemble-median modeled K profiles reproduce the qualitative cross-shore pattern of shoaling amplification and surf zone attenuation across all three energy classes.

Near-zero event-scale correlation. Pearson correlation between modeled and observed K is $r = 0.00$ to 0.05 across energy classes; for the sample sizes available ($n = 34$–40 gauge-event pairs per class), all three values are statistically indistinguishable from zero ($p > 0.05$, 95% CI spanning zero in all cases). The model reproduces the ensemble-mean transformation level but has no detectable event-scale predictive skill against the boundary-referenced metric under the present 1D fixed-bed JONSWAP configuration. Referencing instead to the delivered energy recovers a weak but statistically resolvable relationship (pooled $r = 0.21$, event-block bootstrap 95% CI [0.11, 0.30], $p < 0.001$). The accompanying reductions in bias and RMSE are arithmetic, because K_{bnd} derives from the same run, so only the correlation is informative; it explains about 4% of the pair-to-pair variance, and practical predictive skill remains low. That the recovery survives when K_{bnd} is predicted from the forcing direction alone on withheld events (pooled $r = 0.20$, [0.11, 0.29]) shows it is not an artifact of using the run's own output.

Morphological state. No systematic dependence of model skill on sandbar morphological state is identified. This null result is most likely entailed by the boundary condition limitation: when K_{bnd} scatter dominates the modeled K signal, the model cannot distinguish single-barred from unbarred conditions regardless of its cross-shore physics. The study was not designed to test morphological sensitivity at the required precision and should not be interpreted as evidence that XBeach-NH is morphologically insensitive.

Numerical sensitivity. The $\beta_{\max}$ (breaking steepness threshold) sensitivity tests confirm this is the primary tunable parameter in NH mode, producing the largest responses in the sensitivity package for energetic forcing. Even so, the $\beta_{\max}$ response is substantially smaller than the boundary transmission deficit, and no perturbation altered the central diagnostic conclusion. The roller-model parameters (γ; α, $\gamma_{\max}$) are structurally insensitive in NH mode, as expected. Standard grid adequacy is confirmed for moderate and quiescent events. For the most energetic event it could not be demonstrated: the fine-grid run is numerically unstable and terminates early, so no fine-grid solution exists to compare against (Section 3.6). This is a limitation of depth-averaged

non-hydrostatic models at high forcing, not an outstanding test.

Observational contribution. The K profiles derived from the FRF gauge array document, using a 14-year stratified 52-event design with the permanent gauge array, how cross-shore wave transformation varies across energy classes and sandbar morphological states at Duck. Relative to prior FRF transformation studies, which were largely confined to single field campaigns (Elgar et al., 1997; Gallagher et al., 1998; Ruessink et al., 2001), the novel element here is the combination of a 14-year stratified multi-event design, the normalized cross-class K framework, and the permanent-array sampling. Because the observed K uses the measured offshore H_{m0} as its denominator, the observed profiles are independent of the model and its boundary deficit: the shoaling amplification of 4 to 15% at intermediate positions and the breaking-induced K reduction to a mean of 0.66 at the Paros gauge under energetic forcing are purely observational. These profiles constitute a candidate basis for future model assessment at this site, provided that the boundary transmission behaviour of the candidate model is characterized before absolute comparisons are attempted.

Hydrostatic baseline. A phase-averaged hydrostatic breaking model (Battjes and Janssen, 1978; Thornton and Guza, 1983) was initialized with the same prescribed boundary wave height over the same bathymetry, but is not subject to the wave-generation deficit. It reproduces the observed cross-shore transformation with genuine event-scale skill (Pearson $r = 0.65$, 95% CI 0.57 to 0.73, bias -0.02 across 203 gauge-event pairs), in contrast to the near-zero skill of the 1D non-hydrostatic runs on the identical pairs. This is consistent with the non-hydrostatic skill gap being dominated by boundary energy delivery and not by the breaking physics: once the prescribed boundary energy is supplied, standard depth-limited breaking captures the transformation at this site. The comparison is not on an equal footing, however, because the baseline is driven directly with the prescribed boundary energy and so escapes the transmission deficit by construction; a higher baseline K does not by itself demonstrate superior physics, and a clean attribution would require propagating the same deficient boundary through both a hydrostatic and a non-hydrostatic solver. A complementary three-event in-model surfbeat ablation indicates that the deficit is present in the hydrostatic model as well ($K_{bnd} = 0.53$ to 0.89), so a component of it is a shared property of the 1D directional boundary. The non-hydrostatic deficit is larger in those three cases, but extending the ablation across the event set would be needed to establish that the solver-specific component scales with sea state.

Corrected-boundary controlled test. The controlled comparison that the baseline could not provide, supplying the correct offshore energy to the same non-hydrostatic solver, was run by inflating the boundary forcing by $1/K_{bnd}$ per event and rerunning the full 52-event set. The correction recovers significant event-scale skill: pooled across gauge-event pairs the Pearson correlation rises from 0.04 to 0.30 ([0.16, 0.43], $p = 5.4 \times 10^{-5}$, valid events), with the mean bias contracting from -0.54 to -0.18 and the RMSE from 0.57 to 0.26. This supports the conclusion that boundary energy delivery was the dominant limitation of the uncorrected 1D runs: the inflation factor comes from the runs being corrected, so the experiment is an a posteriori sensitivity test. Predicting the inflation from the prescribed wave direction alone, and scoring only events withheld from the fit, recovers the same skill (pooled $r = 0.20$, 95% CI 0.11 to 0.29). The recovery therefore does not depend on knowledge of the run being corrected. The support is partial. A residual negative bias remains, the recovered correlation explains under one-tenth of the pair-to-pair variance, and the recovery is concentrated in the energetic class. For the seven most energetic events the required inflation drives the boundary H_{m0} past depth-limited breaking, so those events cannot be tested under corrected forcing without a 2DH or deeper-water configuration.

Future work. The normalized assessment framework is reusable for any nearshore wave model applied against the permanent FRF array. Priority directions include: extending the spectral characterization of the boundary deficit, established on the flat bottom in Section 3.12, to the production event set with high-frequency boundary output; testing 2DH or phase-resolving spectral boundary conditions to reduce K_{bnd} toward unity (Rijnsdorp et al., 2015); adding an infragravity-band assessment; and extending the baseline-corrected comparison to a 2DH non-hydrostatic configuration that delivers the prescribed boundary energy while resolving phase-scale and infragravity processes.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.oceaneng.2026.127920.

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